

11.3 ■ THE SOLAR CYCLE

Some of the most fascinating and complex features of the solar atmosphere are transient in nature. However, as we will learn in this section, many observational features of the solar atmosphere are also cyclic.

Sunspots

It was Galileo who made the first telescopic observations of **sunspots** (recall Fig. 11.11). Sunspots are even visible occasionally with the unaided eye, but making such observations is **strongly discouraged** because of the potential for eye damage.

Reliable observations made over the past two centuries indicate that the number of sunspots is approximately periodic, going from minimum to maximum and back to minimum again nearly every 11 years (Fig. 11.28). The average latitude of sunspot formation is also periodic, again over an 11-year cycle. A plot of sunspot location as a function of time is shown in Fig. 11.29, along with a plot of the percentage of the solar surface covered by sunspots. Because of its wing-like appearance, the top portion of Fig. 11.29 has come to be known as the **butterfly diagram**. Individual sunspots are short-lived features, typically surviving no more than a month or so. During its lifetime, a sunspot will remain at a constant latitude, although succeeding sunspots tend to form at progressively lower latitudes. As the last sunspots of one cycle vanish near the Sun's equator, a new cycle begins near $\pm 40^\circ$ (north and south) of the equator. The largest number of spots (sunspot maximum) typically occurs at intermediate latitudes.

The key to understanding sunspots lies in their strong magnetic fields. A typical sunspot is shown in Fig. 11.30. The darkest portion of the sunspot is known as the **umbra** and

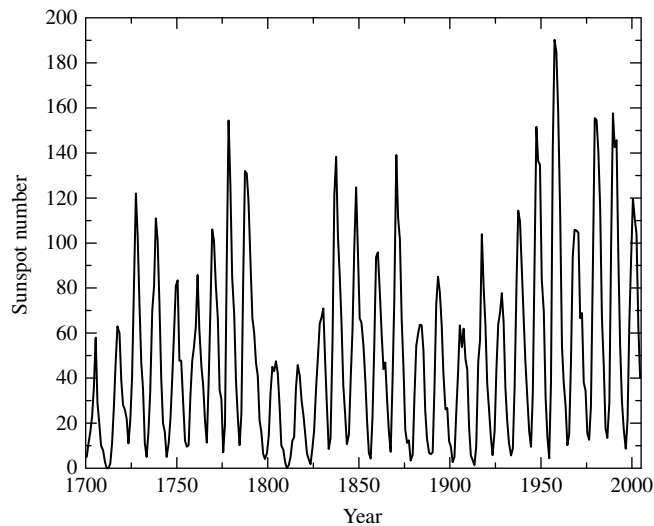


FIGURE 11.28 The number of sunspots between 1700 and 2005 indicates an 11-year periodicity. (Data from the World Data Center for the Sunspot Index at the Royal Observatory of Belgium.)

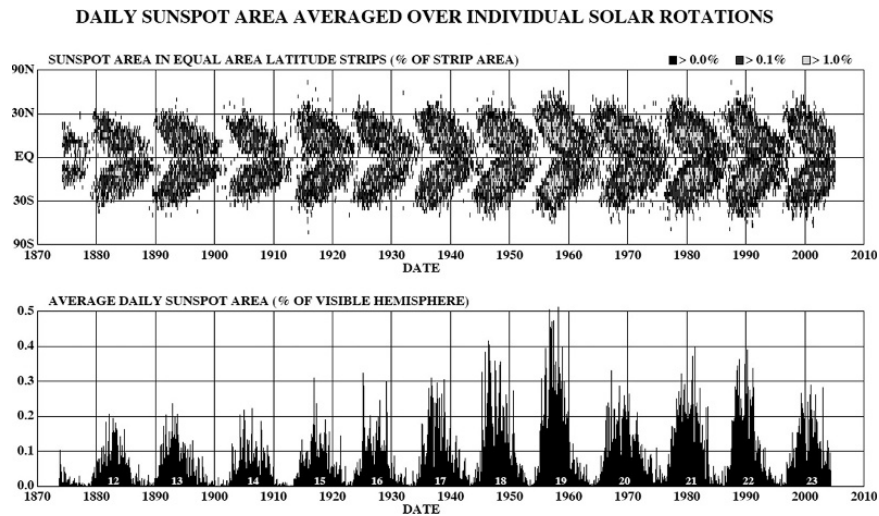


FIGURE 11.29 The upper figure depicts the butterfly diagram, showing sunspot latitudes with time. The lower figure shows the percentage of the Sun's surface covered by sunspots as a function of time. (Courtesy of Dr. David H. Hathaway, NASA/Marshall Space Flight Center.)

may measure as much as 30,000 km in diameter. (For reference, the diameter of Earth is 12,756 km.) The umbra is usually surrounded by a filament-like structure, called the **penumbra**, whose mere appearance suggests the presence of magnetic lines of force. The existence of a strong magnetic field can be verified by observing individual spectral lines produced within the spot. As was discussed in Section 5.4, the strength and polarity of magnetic fields can be measured by observing the Zeeman effect, the splitting of spectral lines that results from removing the degeneracy inherent in atomic energy levels. The amount of splitting is proportional to the strength of the magnetic field, whereas the polarization of the light corresponds to the direction of the field. Figure 11.31 shows an example of the splitting of a spectral line measured across a sunspot. Magnetic field strengths of several tenths of a tesla and greater have been measured in the centers of umbral regions, with field strengths decreasing across penumbral regions. Furthermore, polarization measurements indicate that the direction of a typical umbral magnetic field is vertical, becoming horizontal across the penumbra.

Sunspots are generally located in groups. Typically, a dominant sunspot leads in the direction of rotation, and one or more sunspots follow. During an 11-year cycle, the lead sunspot will always have the same polarity in one hemisphere—say, a north pole in the geographic northern hemisphere—while the lead sunspot in the other hemisphere will have the opposite polarity (e.g., a south pole in the geographic southern hemisphere); trailing sunspots have the opposite polarity. Even when a large collection of trailing spots exist, resulting from a tangled magnetic field pattern, a basically *bipolar* field is present. During the next 11-year cycle, polarities will be reversed; the sunspot with a magnetic south polarity will lead in the northern hemisphere, and vice versa in the southern hemisphere. Accompanying this local polarity reversal is a global polarity reversal: the overall dipole field of the Sun

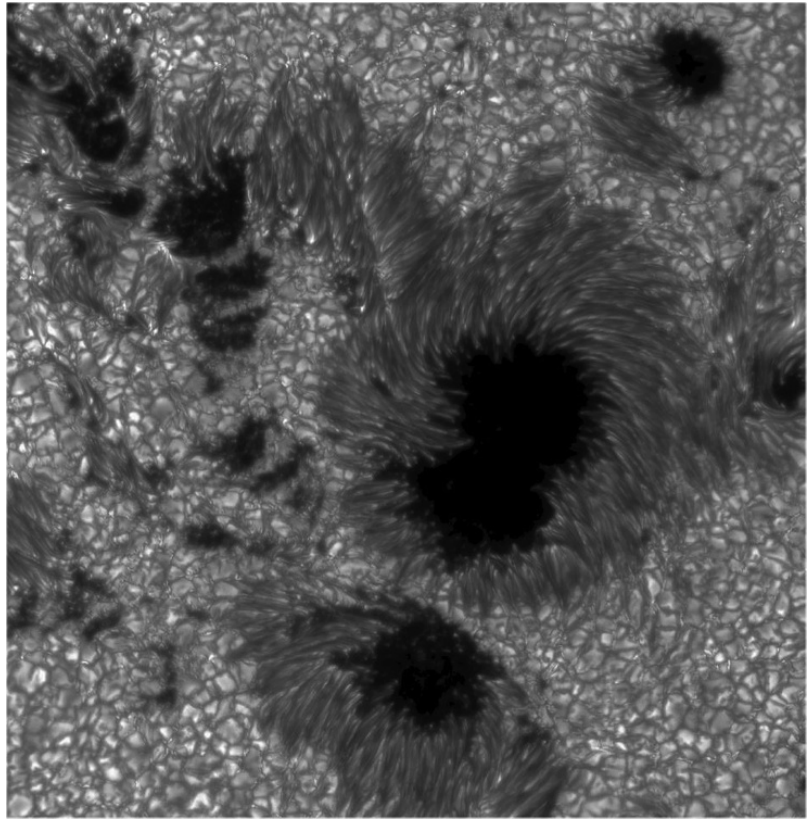


FIGURE 11.30 A typical sunspot group. The dark umbra of the central sunspot is clearly evident, as is the filamentary structure of its penumbra. (Courtesy of the Royal Swedish Academy of Sciences.)

will change so that the magnetic north pole of the Sun will switch from the geographic north pole to the geographic south pole. Polarity reversal always occurs during sunspot minimum, when the first sunspots are beginning to form at the highest latitudes. When the polarity reversal is considered, the Sun is said to have a **22-year cycle**. This important magnetic behavior is illustrated in Fig. 11.32.

The dark appearance of sunspots is due to their significantly lower temperatures. In the central portion of the umbra the temperature may be as low as 3900 K, compared with the Sun's effective temperature of 5777 K. From Eq. (3.18), this implies a surface bolometric flux that is a factor of $(5777/3900)^4 = 4.8$ lower than that of the surrounding photosphere.¹³ Observations obtained from the Solar Maximum Mission satellite (SMM) have shown that this decrease in surface flux affects the overall energy output of the Sun. When a number of large sunspots exist, the solar luminosity is depressed by roughly 0.1%. Since convection is the principal energy transport mechanism just below the photosphere, and since strong

¹³A 3900-K blackbody is very bright, of course. However, when seen through a filter dark enough to make viewing the rest of the 5777-K photosphere comfortable, the sunspot appears dark.

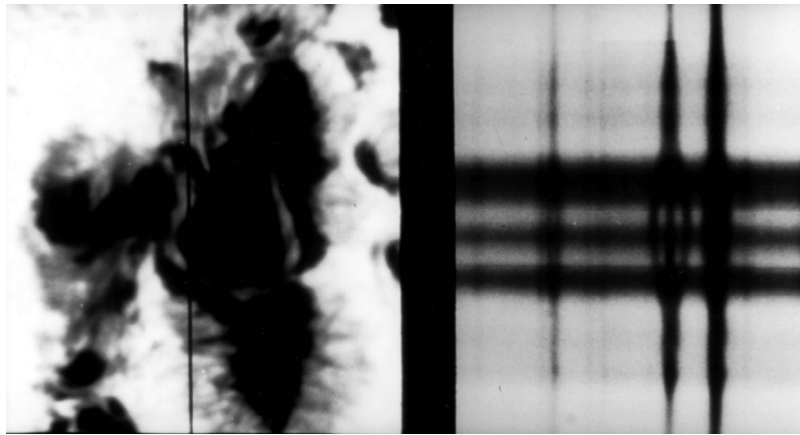


FIGURE 11.31 The Zeeman splitting of the Fe 525.02-nm spectral line due to the presence of a strong magnetic field in a sunspot. The spectrograph slit was aligned vertically across a sunspot, resulting in a wavelength dependence that runs from left to right in the image. The slit extended beyond the image of the sunspot. (Courtesy of the National Optical Astronomy Observatories/National Solar Observatory.)

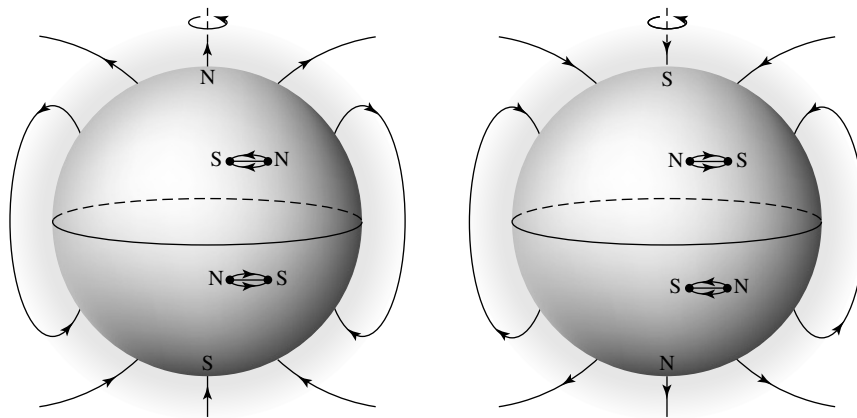


FIGURE 11.32 The global magnetic field orientation of the Sun, along with the magnetic polarity of sunspots during successive 11-year periods.

magnetic fields inhibit motion through the “freezing in” of field lines in a plasma, it is likely that the mass motion of convective bubbles is inhibited in sunspots, thereby decreasing the flow of energy through the sunspots.

Along with luminosity variations on a timescale of months (the typical lifetime of an individual sunspot), the Sun’s luminosity seems to experience variability on a much longer timescale, as does the number of sunspots. For instance, very few sunspots were observed between 1645 and 1715; this time interval has come to be called the **Maunder minimum**

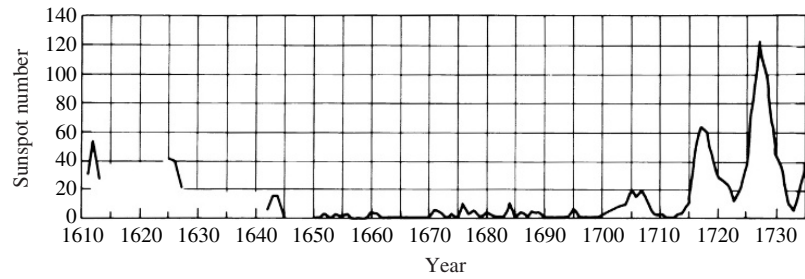


FIGURE 11.33 An unusually small number of sunspots were observed between 1645 and 1715 (the Maunder minimum). (Adapted from a figure courtesy of J. A. Eddy, High Altitude Observatory.)

(see Fig. 11.33).¹⁴ Surprisingly, during this period the average temperature in Europe was significantly lower, consistent with the solar luminosity being a few tenths of a percent less than it is today. John Eddy has proposed that there is a very long-term periodicity on which the solar cycle is superimposed. This long-period variation goes through grand sunspot maxima and minima that may last for centuries. Evidence in support of this suggestion is found on Earth in the relative numbers of atmospheric carbon dioxide molecules that contain radioactive carbon atoms (^{14}C), as preserved in the 7000-year-long record of tree rings. The importance of ^{14}C in long-term sunspot studies lies in an inverse correlation between sunspots and the amount of ^{14}C present in Earth's atmosphere. ^{14}C is a radioactive isotope of carbon that is produced when extremely energetic charged particles from space, called **cosmic rays**, collide with atmospheric nitrogen. Cosmic rays are affected by the magnetic field of the Sun, which in turn is affected by solar activity. During the Maunder minimum, the amount of atmospheric ^{14}C increased significantly and was incorporated into the rings of living trees. The amount of ^{14}C also seems to correlate well with the advance and retreat of glaciers over the past 5000 years.

With lower temperatures in sunspots, the gas pressure is necessarily lower than in the surrounding material (see Eq. 10.11). However, the gravitational force is essentially the same. From these considerations alone, it seems as though the gas within a sunspot ought to sink into the interior of the star, an effect that is not observed. Without the benefit of a sufficiently large gas pressure gradient to support a sunspot, another component to the pressure must exist. As we have already seen in the last section, a magnetic field is accompanied by a pressure term. It is this extra magnetic pressure that provides the support necessary to keep a sunspot from sinking or being compressed by the surrounding gas pressure.

Plages

A variety of other phenomena are also associated with sunspot activity. **Plages** (from the French word for *beach*) are chromospheric regions of bright $\text{H}\alpha$ emission located near

¹⁴With the development and continual improvement of the telescope beginning during the early phase of the Maunder minimum (recall that Galileo died in 1642 and Newton was born in the same year), the Maunder minimum was not a manifestation of poor observations.

active sunspots (recall Fig. 11.17). They usually form before the sunspots appear and usually disappear after the sunspots vanish from a particular area. Plages have higher densities than the surrounding gas and are products of the magnetic fields. Apparently the cause of the decreased brightness of sunspots does not play an important role in plages.

Solar Flares

Solar flares are eruptive events that are known to release from 10^{17} J of energy at the lower detection limit to as much as 10^{25} J of energy over time intervals ranging from milliseconds to more than an hour.¹⁵ The physical dimensions of a flare are enormous, with a large flare reaching 100,000 km in length (see Fig. 11.34a). During an eruption, the hydrogen Balmer line, $H\alpha$, appears locally in emission rather than in absorption, as is usually the case, implying that photon production occurs above much of the absorbing material. When observed in $H\alpha$, a flare is often seen on the disk as two ribbons of light (Fig. 11.34b). Along with $H\alpha$, other types of electromagnetic radiation are produced that can range from kilometer-wavelength *nonthermal* radio waves due to synchrotron radiation (see Section 4.3) to very short-wavelength hard X-ray and gamma-ray emission lines.

Charged particles are also ejected outward at high speeds, many escaping into interplanetary space as **solar cosmic rays**. In the largest flares the ejected charged particles, mostly protons and helium nuclei, may reach Earth in 30 minutes, disrupting some communications and posing a very serious threat to any unprotected astronauts. Shock waves are also generated and can occasionally propagate several astronomical units before dissipating.

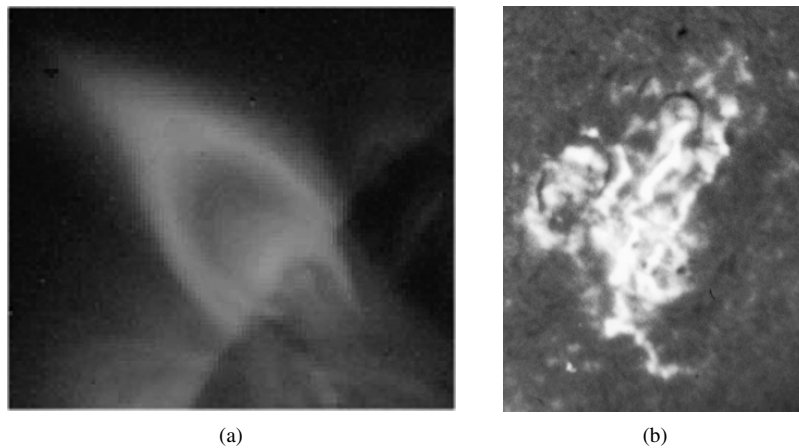


FIGURE 11.34 (a) A solar flare seen at the limb of the Sun, observed by the Yohkoh Soft X-ray Telescope, March 18, 1999, 16:40 UT. (From the Yohkoh mission of ISAS, Japan. The X-ray telescope was prepared by the Lockheed Palo Alto Research Laboratory, the National Astronomical Observatory of Japan, and the University of Tokyo with the support of NASA and ISAS.) (b) A two-ribbon flare seen in $H\alpha$ on October 19, 1989. (Courtesy of the National Optical Astronomy Observatories.)

¹⁵For comparison, a one-megaton bomb releases approximately 10^{16} J.

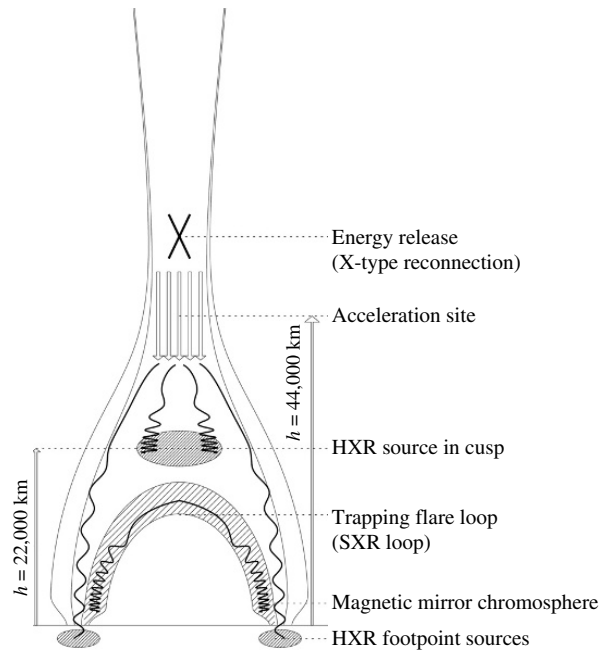


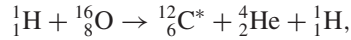
FIGURE 11.35 A model of the January 13, 1992, Masuda solar flare. Note the two hard X-ray (HXR) footpoint sources associated with $H\alpha$ flare ribbons [see Fig. 11.34(b)]. Electrons are accelerated downward along the magnetic field lines until they collide with the chromosphere. The soft X-ray (SXR) loop may be compared to Fig. 11.34(a). (Figure adapted from Aschwanden, et al., *Ap. J.*, 464, 985, 1996.)

The answer to the question of what powers solar flares lies in the location of the flare eruption. Flares develop in regions where the magnetic field intensity is great, namely in sunspot groups. From the discussion of the previous section, the creation of magnetic fields results in energy being stored in those magnetic fields (Eq. 11.9). If a magnetic field disturbance could quickly release the stored energy, a flare might develop. It is left as an exercise to show that both the amount of energy stored in the magnetic field and the timescale involved in perturbing it via Alfvén waves are consistent with the creation of a solar flare. However, details of the energy conversion, such as particle acceleration, are still a matter of active research.

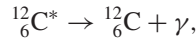
A model of a solar flare is illustrated in Fig. 11.35. The general mechanism of a solar flare involves the **reconnection** of magnetic field lines. A disturbance in magnetic field loops (perhaps due to the Sun's convection zone) causes the creation of a sheet of current in the highly conducting plasma (recall Lenz's law). The finite resistance in the plasma results in Joule heating of the gas, causing temperatures to reach 10^7 K. Particles accelerated away from the reconnection point and away from the Sun may escape entirely, producing solar cosmic rays. Radio-wavelength radiation is generated by the synchrotron process of charged particles spiraling around the magnetic field lines. Soft X-ray emission results from the high temperatures in the loop below the acceleration (reconnection) point. $H\alpha$ emission at the base of the magnetic field lines (the two $H\alpha$ ribbons) is produced by recombining

electrons and protons that are accelerated away from the reconnection point, toward the chromosphere.

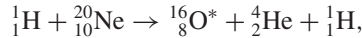
In addition, high-energy particles accelerated toward the chromosphere produce hard X-rays and gamma rays due to surface nuclear reactions. Examples of important nuclear reactions associated with solar flares are **spallation reactions** that break heavier nuclei into lighter nuclei, such as



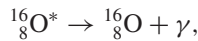
where C^* represents a carbon nucleus in an excited state, followed by the de-excitation reaction



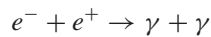
with $E_\gamma = 4.438 \text{ MeV}$, or



followed by the de-excitation reaction



with $E_\gamma = 6.129 \text{ MeV}$. Other examples of reactions produced by flares on the Sun's surface include electron-positron annihilation,



where $E_\gamma = 0.511 \text{ MeV}$, and the production of deuterium by

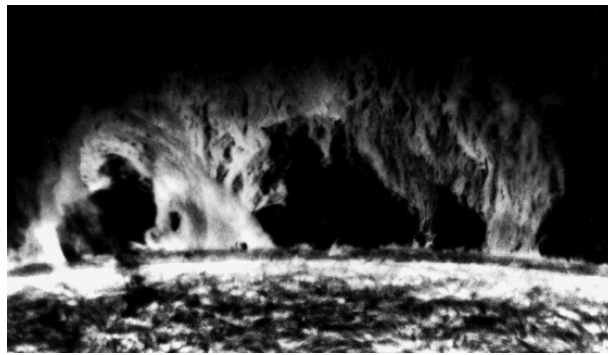


where $E_\gamma = 2.223 \text{ MeV}$.

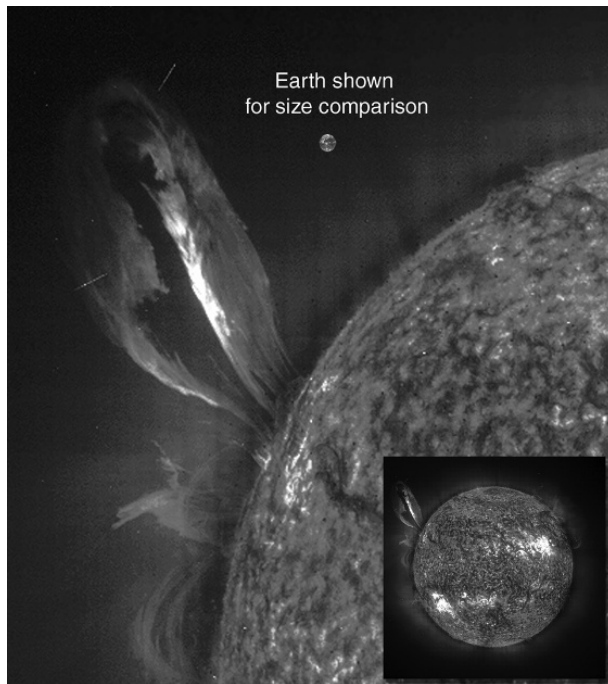
Solar Prominences

Solar prominences are also related to the Sun's magnetic field. **Quiescent prominences** are curtains of ionized gas that reach well into the corona and can remain stable for weeks or months. The material in the prominence has collected along the magnetic field lines of an active region, with the result that the gas is cooler (with a typical temperature of 8000 K) and more dense than the surrounding coronal gas. This causes the gas to "rain" back down into the chromosphere. When viewed in $\text{H}\alpha$ at the limb of the Sun, quiescent prominences appear as bright structures against the thin corona. However, when viewed in the continuum against the solar disk, a quiescent prominence appears as a dark **filament**, absorbing the light emitted from below. An example of a quiescent prominence is shown in Fig. 11.36(a).

An **eruptive** (or **active**) **prominence** (Fig. 11.36b) may exist for only a few hours and may abruptly develop from a quiescent prominence. It appears as though a relatively stable magnetic field configuration can suddenly become unstable, causing the prominence to lift away from the Sun. Although the mechanism is related to that of a solar flare, the outcome is somewhat different; rather than most of the energy going into electromagnetic radiation, the energy of an eruptive prominence is converted into mass motions as gas is ejected from the Sun.



(a)



(b)

FIGURE 11.36 (a) A quiescent hedgerow prominence. (Courtesy of Big Bear Solar Observatory, California Institute of Technology.) (b) An eruptive prominence observed by the SOHO Extreme Ultraviolet Imaging Telescope (EIT) on July 24, 1999. [SOHO (ESA & NASA)]

Coronal Mass Ejections

Even more spectacular is a **coronal mass ejection** (CME). CMEs have been observed since the early 1970s using spacecraft such as NASA's seventh Orbiting Solar Observatory (OSO 7) and Skylab. Most recently, CMEs have been observed routinely by SOHO's Large Angle Spectrometric COronagraph (LASCO); see Fig. 11.37. LASCO uses an occulting disk to create an artificial solar eclipse, allowing it to observe the white-light corona from

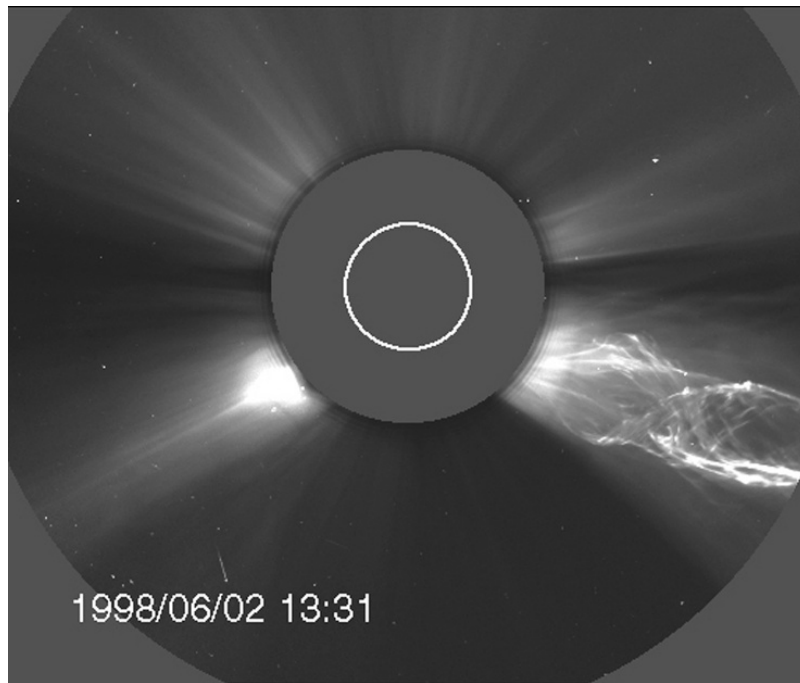


FIGURE 11.37 A coronal mass ejection observed by the SOHO LASCO instrument on June 2, 1998. Note the intricacy of the magnetic field lines within the CME. The white circle on the occulting disk represents the size of the Sun out to the photosphere. [SOHO (ESA & NASA)]

a few solar radii out to $30 R_{\odot}$. With the detection of thousands of CMEs it appears that there is about one CME per day when averaged over the 11-year sunspot cycle. When the Sun is more active (i.e., near sunspot maximum) the frequency may be about 3.5 events per day, and during sunspot minimum the number of events may decrease to roughly one every five days. During a CME event, between 5×10^{12} kg and 5×10^{13} kg of material may be ejected from the Sun at speeds ranging from 400 km s^{-1} to over 1000 km s^{-1} . CMEs appear to be associated with eruptive prominences approximately 70% of the time, and with flares only about 40% of the time. One can think of a CME as a magnetic bubble lifting off of the Sun's surface after a magnetic reconnection event, carrying a significant fraction of the mass of the solar corona with it.

The Time-Dependent Shape of the Corona

Yet another feature of the solar cycle involves the shape of the corona itself. During a period of little solar activity, when there are few sunspots and few, if any, flares or prominences, the **quiet corona** is generally more extended at the equator than at the poles, consistent with a nearly dipole magnetic field. Near sunspot maximum, the **active corona** is more complex in shape, as is the structure of the magnetic field. Examples of the shape of the corona during sunspot minimum and maximum are seen in Figs. 11.20(a) and 11.20(b), respectively. Evidently, the changing shape of the corona, like other solar activity, is due to the dynamic structure of the Sun's magnetic field.

The Magnetic Dynamo Theory

A **magnetic dynamo** model describing many of the components of the solar cycle was first proposed by Horace Babcock in 1961. Despite its general success in describing the major features of the solar cycle, the model is not yet able to provide adequate explanations of many of the important details of solar activity. Any complete picture of the solar cycle will require a full treatment of the MHD equations in the solar environment, including differing rotation rates with latitude and depth in the Sun, convection, solar oscillations, heating of the upper atmosphere, and mass loss. Of course, not all of these processes are likely to play equally important roles in the study of the solar cycle, but it is important to understand the degree to which each of them contributes to the particular phenomenon under investigation.

As depicted in Fig. 11.38, because the magnetic field lines are “frozen into” the gas, the differential rotation of the Sun drags the lines along, converting a *poloidal* field (essentially

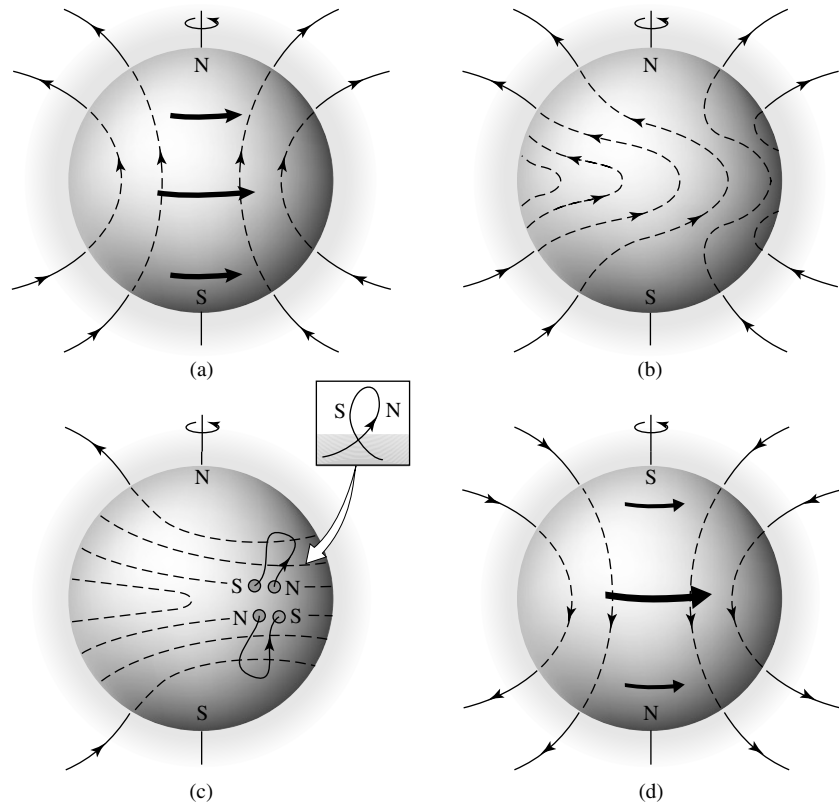


FIGURE 11.38 The magnetic dynamo model of the solar cycle. (a) The solar magnetic field is initially a poloidal field. (b) Differential rotation drags the “frozen-in” magnetic field lines around the Sun, converting the poloidal field into a toroidal field. (c) Turbulent convection twists the field lines into magnetic ropes, causing them to rise to the surface as sunspots, the polarity of the lead spots corresponding to the original polarity of the poloidal field. (d) As the cycle progresses, successive sunspot groups migrate toward the equator where magnetic field reconnection reestablishes the poloidal field, but with the original polarity reversed.

a simple magnetic dipole) to one that has a significant *toroidal* component (field lines that are wrapped around the Sun). The turbulent convection zone then has the effect of twisting the lines, creating regions of intense magnetic fields, called magnetic *ropes*. The buoyancy produced by magnetic pressure (Eq. 11.10) causes the ropes to rise to the surface, appearing as sunspot groups. The polarity of the sunspots is due to the direction of the magnetic field along the ropes; consequently, every lead spot in one hemisphere will have the same polarity while the lead spots in the other hemisphere will have the opposite polarity.

Initially, the little twisting that does develop occurs at higher latitudes; during sunspot minimum. As the differential rotation continues to drag the field lines along and convective turbulence ties them in knots, more sunspots develop at intermediate latitudes, producing a sunspot maximum. It would seem that ultimately the greatest amount of twisting and the largest number of sunspots should develop near the equator. However, sunspots from the two hemispheres tend to cancel out near the equator since the polarities of their leading spots are opposed. As a result, the number of sunspots appearing near the equator is small. Finally, the cancelation of magnetic fields near the equator causes the poloidal field to be reestablished, but with its original polarity reversed. This process takes approximately 11 years. The entire procedure repeats continuously, with the polarity of the magnetic field returning to its original orientation every other cycle. Hence, the entire solar cycle is actually 22 years long when magnetic field polarities are considered.

As we have already seen, details related to specific phenomena, such as the cause of the decreased flux coming from sunspots or the exact process of flare generation, are not yet well understood. The same situation also holds for the more fundamental magnetic dynamo itself. Although the preceding discussion describes the behavior of the solar cycle in an approximate way, even such basic results as the timescales involved have not yet been accurately modeled. A successful magnetic dynamo model must not only produce the general location and numbers of sunspots and flares, but it must also do so with the observed 22-year periodicity. Moreover, the dynamo model must replicate the much slower variation that appears to be responsible for the Maunder minimum.

Evidence of Magnetic Activity in Other Stars

Fortunately, some evidence does exist that the basic ideas behind the solar cycle are correct. Observations of other cool main-sequence stars indicate that they possess activity cycles much like the solar cycle. It was pointed out in the last section that late main-sequence stars exhibit observational characteristics consistent with the existence of hot coronae. It was also mentioned that angular momentum is apparently lost via stellar winds. Both phenomena agree with the theoretical onset of surface convection in low-mass stars, a major component of the dynamo theory.

Other forms of magnetic activity have also been seen in some stars. Observations indicate the existence of **flare stars**, main-sequence stars of class M that demonstrate occasional, rapid fluctuations in brightness. If flares the size of those on the Sun were to occur on the much dimmer M stars, the flares would contribute significantly to the total luminosity of those stars, producing the short-term changes that are observed. Much larger flares may also be generated by other stars as well: On April 24, 2004, the star GJ 3685A released a flare that was roughly one million times more energetic than a large solar flare. The event was detected serendipitously by NASA's Galaxy Evolution Explorer.

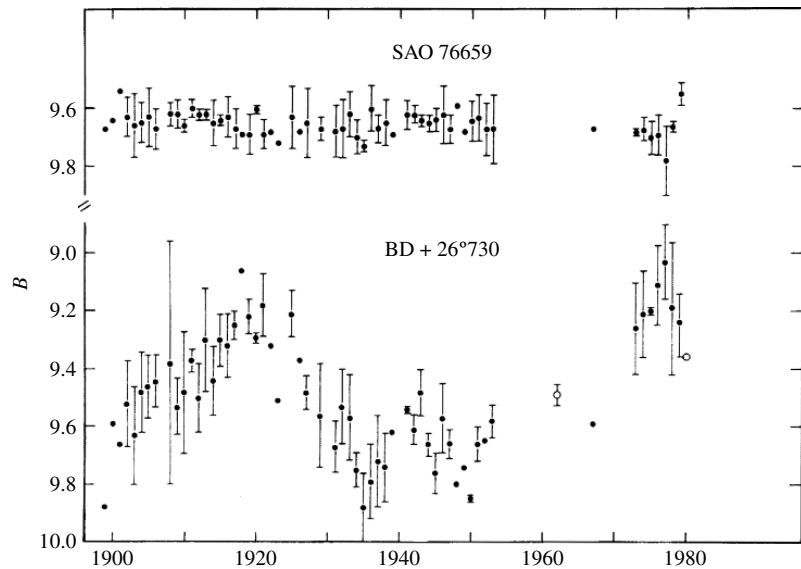


FIGURE 11.39 The light curve of BD + 26°730, a BY Dra star. SAO 76659 is a nearby reference star. (Figure from Hartmann et al., *Ap. J.*, 249, 662, 1981.)

Starspots are also observed to exist on stars other than the Sun. Starspots are revealed by their effect on the luminosity of a star, which can be measured at a level of 1%. Two classes of stars, RS Canum Venaticorum and BY Draconis stars,¹⁶ show significant long-term variations that are attributed to starspots covering appreciable fractions of their surfaces. For example, Fig. 11.39 shows a variation of over 0.6 magnitude in the B band for the BY Draconis star, BD + 26°730. Starspots can even be used to measure stellar rotation.

Magnetic fields have also been detected directly on several cool main-sequence stars by measuring Zeeman-broadened spectral lines. Analysis of the data indicates field strengths of several tenths of a tesla over significant fractions of the stellar surfaces. The existence of the strong fields correlates with their observed luminosity variations.

From our discussion in this chapter, it should be clear that astrophysics has had a great deal of success in explaining many of the features of our Sun. The stellar structure equations describe the major aspects of the solar interior, and much of the Sun's complex atmosphere is also understood. But many other important issues remain to be resolved, such as the surface abundance of lithium, the intricate details of the solar cycle, and the interaction between the Sun and Earth's climate. Much exciting and challenging work remains to be done before we can feel confident that we fully understand the star that is closest to us.

¹⁶Classes of stars that show light variations, *variable stars*, are usually named after the first star discovered that exhibits the specific characteristics. RS CVn and BY Dra are main-sequence stars of spectral classes F–G and K–M, respectively. The letters RS and BY indicate that these are variable stars; Canum Venaticorum and Draconis are the constellations in which the stars are located.

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PROBLEMS

- 11.1** Using Fig. 11.1, verify that the change in the Sun's effective temperature over the past 4.57 billion years is consistent with the variations in its radius and luminosity.
- 11.2** (a) At what rate is the Sun's mass decreasing due to nuclear reactions? Express your answer in solar masses per year.
- (b) Compare your answer to part (a) with the mass loss rate due to the solar wind.
- (c) Assuming that the solar wind mass loss rate remains constant, would either mass loss process significantly affect the total mass of the Sun over its entire main-sequence lifetime?
- 11.3** Using the Saha equation, calculate the ratio of the number of H^- ions to neutral hydrogen atoms in the Sun's photosphere. Take the temperature of the gas to be the effective temperature, and assume that the electron pressure is 1.5 N m^{-2} . Note that the Pauli exclusion principle requires that only one state can exist for the ion because its two electrons must have opposite spins.
- 11.4** The Paschen series of hydrogen ($n = 3$) can contribute to the visible continuum for the Sun since the series limit occurs at 820.8 nm. However, it is the contribution from the H^- ion that dominates the formation of the continuum. Using the results of Problem 11.3, along with the Boltzmann equation, estimate the ratio of the number of H^- ions to hydrogen atoms in the $n = 3$ state.
- 11.5** (a) Using Eq. (9.63) and neglecting turbulence, estimate the full width at half-maximum of the hydrogen $H\alpha$ absorption line due to random thermal motions in the Sun's photosphere. Assume that the temperature is the Sun's effective temperature.
- (b) Using $H\alpha$ redshift data for solar granulation, estimate the full width at half-maximum when convective turbulent motions are included with thermal motions.
- (c) What is the ratio of v_{turb}^2 to $2kT/m$?
- (d) Determine the relative change in the full width at half-maximum due to Doppler broadening when turbulence is included. Does turbulence make a significant contribution to $(\Delta\lambda)_{1/2}$ in the solar photosphere?

- 11.6** Estimate the thermally Doppler-broadened line widths for the hydrogen Ly α , C III, O VI, and Mg X lines given on page 365; use the temperatures provided. Take the masses of H, C, O, and Mg to be 1 u, 12 u, 16 u, and 24 u, respectively.

- 11.7 (a)** Using Eq. (3.22), show that in the Sun's photosphere,

$$\ln (B_a/B_b) \approx 11.5 + \frac{hc}{kT} \left(\frac{1}{\lambda_b} - \frac{1}{\lambda_a} \right)$$

where B_a/B_b is the ratio of the amount of blackbody radiation emitted at $\lambda_a = 10$ nm to the amount emitted at $\lambda_b = 100$ nm, centered in a wavelength band 0.1 nm wide.

- (b) What is the value of this expression for the case where the temperature is taken to be the effective temperature of the Sun?
- (c) Writing the ratio in the form $B_a/B_b = 10^x$, determine the value of x .
- 11.8** The gas pressure at the base of the photosphere is approximately 2×10^4 N m $^{-2}$ and the mass density is 3.2×10^{-4} kg m $^{-3}$. Estimate the sound speed at the base of the photosphere, and compare your answer with the values at the top of the photosphere and averaged throughout the Sun.
- 11.9** Suppose that you are attempting to make observations through an optically thick gas that has a constant density and temperature. Assume that the density and temperature of the gas are 2.2×10^{-4} kg m $^{-3}$ and 5777 K, respectively, typical of the values found in the Sun's photosphere. If the opacity of the gas at one wavelength (λ_1) is $\kappa_{\lambda 1} = 0.026$ m 2 kg $^{-1}$ and the opacity at another wavelength (λ_2) is $\kappa_{\lambda 2} = 0.030$ m 2 kg $^{-1}$, calculate the distance into the gas where the optical depth equals 2/3 for each wavelength. At which wavelength can you see farther into the gas? How much farther? This effect allows astronomers to probe the Sun's atmosphere at different depths (recall Fig. 11.13).
- 11.10 (a)** Using the data given in Example 11.2.2, estimate the pressure scale height at the base of the photosphere.
- (b) Assuming that the ratio of the mixing length to the pressure scale height is 2.2, use the measured Doppler velocity of solar granulation to estimate the amount of time required for a convective bubble to travel one mixing length. Compare this value to the characteristic lifetime of a granule.
- 11.11** Show that Eq. (11.6) follows directly from Eq. (11.5).
- 11.12** Calculate the magnetic pressure in the center of the umbra of a large sunspot. Assume that the magnetic field strength is 0.2 T. Compare your answer with a typical value of 2×10^4 N m $^{-2}$ for the gas pressure at the base of the photosphere.
- 11.13** Assume that a large solar flare erupts in a region where the magnetic field strength is 0.03 T and that it releases 10^{25} J in one hour.
- (a) What was the magnetic energy density in that region before the eruption began?
- (b) What minimum volume would be required to supply the magnetic energy necessary to fuel the flare?
- (c) Assuming for simplicity that the volume involved in supplying the energy for the flare eruption was a cube, compare the length of one side of the cube with the typical size of a large flare.
- (d) How long would it take an Alfvén wave to travel the length of the flare?

- (e) What can you conclude about the assumption that magnetic energy is the source of solar flares, given the physical dimensions and timescales involved?
- 11.14** Assuming that an average of one coronal mass ejection occurs per day and that a typical CME ejects 10^{13} kg of material, estimate the annual mass loss from CMEs and compare your answer with the annual mass loss from the solar wind. Express your answer as a percentage of CME mass loss to solar wind mass loss.
- 11.15** Assume that the velocity of a CME directed toward Earth is 400 km s^{-1} and that the mass of the CME is 10^{13} kg.
- (a) Estimate the kinetic energy contained in the CME, and compare your answer to the energy released in a large flare. Express your answer as a percentage of the energy of the flare.
 - (b) Estimate the transit time for the CME to reach Earth.
 - (c) Briefly explain how astronomers are able to “predict” the occurrence of aurorae in advance of magnetic storms on Earth.
- 11.16** (a) Calculate the frequency shift produced by the normal Zeeman effect in the center of a sunspot that has a magnetic field strength of 0.3 T.
- (b) By what fraction would the wavelength of one component of the 630.25-nm Fe I spectral line change as a consequence of a magnetic field of 0.3 T?
- 11.17** From the data given in Fig. 11.16, estimate the rotation period of the solar interior at the base of the tachocline.
- 11.18** Argue from Eq. (11.9) and the work integral that magnetic pressure is given by Eq. (11.10).

CHAPTER

12

The Interstellar Medium and Star Formation

- 12.1 *Interstellar Dust and Gas*
- 12.2 *The Formation of Protostars*
- 12.3 *Pre-Main-Sequence Evolution*

12.1 ■ INTERSTELLAR DUST AND GAS

When we look into the heavens, it appears as though the stars are unchanging, point-like sources of light that shine steadily. On casual inspection, even our own Sun appears constant. But, as we have seen in the last chapter, this is not the case; sunspots come and go, flares erupt, significant amounts of matter are launched into space via coronal mass ejections, the corona itself changes shape, and even the Sun's luminosity appears to be fluctuating over human timescales, as evidenced by the Maunder minimum. Of course, over the 4.57-Gyr lifetime of our Sun, the luminosity, effective temperature, and radius have all changed substantially (recall Fig. 11.1).

In fact all stars change. Usually the changes are so gradual and over such long time intervals when measured in human terms that we do not notice them without very careful telescopic observation. Occasionally, however, the changes are extremely rapid and dramatic, as in the case of a supernova explosion. By invoking the understanding we developed thus far of the physics of stellar interiors and atmospheres, we can now begin to examine the processes governing how stars evolve during their lives.

The Interstellar Medium

In some sense the evolution of stars is a cyclic process. A star is born out of gas and dust that exists between the stars, known as the **interstellar medium** (ISM). During its lifetime, depending on the star's total mass, much of that material may be returned to the ISM through stellar winds and explosive events. Subsequent generations of stars can then form from this processed material. As a result, to understand the evolution of a star, it is important to study the nature of the ISM.

Understanding the interstellar medium is critical for more than its role in stellar evolution, however. The ISM is of profound importance in describing the structure, dynamics, and evolution of our Milky Way Galaxy, as well as galaxies throughout the universe. In addition, it impacts our observations of everything from relatively nearby stars to the most remote galaxies and quasars.

More fundamentally, the ISM is an enormous and complex environment that provides an important laboratory for testing our understanding of astrophysics at many levels. The dynamics of the ISM involve turbulent gas motions, shocks, and galactic magnetic fields that lace through interstellar space. Thus, modeling the ISM ultimately requires detailed solutions to the equations of magnetohydrodynamics. The dust, molecules, atoms, ions, and free electrons that permeate the ISM challenge our understanding of radiative transfer, thermodynamics, and quantum mechanics. Moreover, the production and destruction of dust grains and complex molecules requires a detailed understanding of chemistry in an environment not reproducible in a terrestrial laboratory.

As an introduction to astrophysical processes, this text is unable to explore all of the fascinating aspects of the interstellar medium. Consequently, the present section serves only as a brief introduction to general aspects of the ISM.

Interstellar Extinction

On a dark night some of the **dust clouds** that populate our Milky Way Galaxy can be seen in the band of stars that is the disk of the Galaxy (see Fig. 12.1). It is not that these dark regions are devoid of stars, but rather that the stars located behind intervening dust clouds are obscured. This obscuration, referred to as **interstellar extinction**, is due to the summative effects of scattering and absorption of starlight (as depicted in Fig. 12.2).

Given the effect that extinction can have on the apparent magnitude of a star, the distance modulus equation (Eq. 3.6) must be modified appropriately. In a given wavelength band centered on λ , we now have

$$m_{\lambda} = M_{\lambda} + 5 \log_{10} d - 5 + A_{\lambda}, \quad (12.1)$$

where d is the distance in pc and $A_{\lambda} > 0$ represents the number of magnitudes of interstellar extinction present along the line of sight. If A_{λ} is large enough, a star that would otherwise be visible to the naked eye or through a telescope could no longer be detected. This is the reason for the dark bands running through the Milky Way.

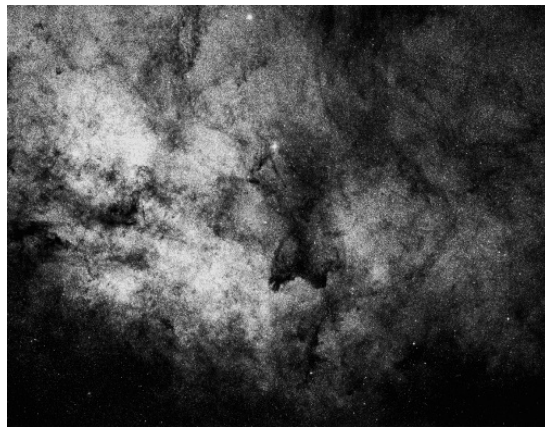


FIGURE 12.1 Dust clouds obscure the stars located behind them in the disk of the Milky Way. (Courtesy of Palomar/Caltech.)

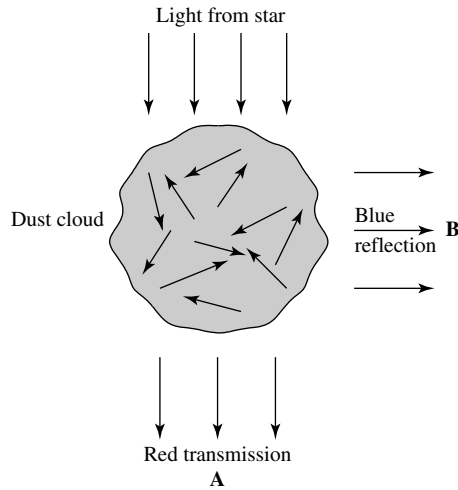


FIGURE 12.2 An interstellar cloud containing significant amounts of dust along with the gas (a dust cloud) can both scatter and absorb light that passes through it. The amount of scattering and absorption depends on the number density of dust grains, the wavelength of the light, and the thickness of the cloud. Since shorter wavelengths are affected more significantly than longer ones, a star lying behind the cloud appears reddened to observer A. Observer B sees the scattered shorter wavelengths as a blue reflection nebula.

Clearly A_λ must be related to the optical depth of the material, measured back along the line of sight. From Eq. (9.18), the fractional change in the intensity of the light is given by

$$I_\lambda / I_{\lambda,0} = e^{-\tau_\lambda},$$

where $I_{\lambda,0}$ is the intensity in the absence of interstellar extinction. Combining this with Eq. (3.4), we can now relate the optical depth to the change in apparent magnitude due to extinction, giving

$$m_\lambda - m_{\lambda,0} = -2.5 \log_{10} (e^{-\tau_\lambda}) = 2.5 \tau_\lambda \log_{10} e = 1.086 \tau_\lambda.$$

But the change in apparent magnitude is just A_λ , so

$$A_\lambda = 1.086 \tau_\lambda. \quad (12.2)$$

The change in magnitude due to extinction is approximately equal to the optical depth along the line of sight.

From the expression for the optical depth given by Eq. (9.17) and the discussion at the end of Example 9.2.2, the optical depth through the cloud is given by

$$\tau_\lambda = \int_0^s n_d(s') \sigma_\lambda ds', \quad (12.3)$$

where $n_d(s')$ is the number density of scattering dust grains and σ_λ is the scattering cross

section. If σ_λ is constant along the line of sight, then

$$\tau_\lambda = \sigma_\lambda \int_0^s n_d(s') ds' = \sigma_\lambda N_d, \quad (12.4)$$

where N_d , the dust grain *column density*, is the number of scattering dust particles in a thin cylinder with a cross section of 1 m^2 stretching from the observer to the star. Thus we see that the amount of extinction depends on the amount of interstellar dust that the light passes through, as one would expect.

The Mie Theory

If we assume for simplicity, as was first done by Gustav Mie (1868–1957) in 1908, that dust particles are spherical and each has a radius a , then the geometrical cross section that a particle presents to a passing photon is just $\sigma_g = \pi a^2$. We may now define the dimensionless **extinction coefficient** Q_λ to be

$$Q_\lambda \equiv \frac{\sigma_\lambda}{\sigma_g},$$

where Q_λ depends on the composition of the dust grains.

Mie was able to show that when the wavelength of the light is on the order of the size of the dust grains, then $Q_\lambda \sim a/\lambda$, implying that

$$\sigma_\lambda \propto \frac{a^3}{\lambda} \quad (\lambda \gtrsim a). \quad (12.5)$$

In the limit that λ becomes very large relative to a , Q_λ goes to zero. On the other hand, if λ becomes very small relative to a , it can be shown that Q_λ approaches a constant, independent of λ so that

$$\sigma_\lambda \propto a^2 \quad (\lambda \ll a). \quad (12.6)$$

These limiting behaviors can be understood by analogy to waves on the surface of a lake. If the wavelength of the waves is much larger than an object in their way, such as a grain of sand, the waves pass by almost completely unaffected ($\sigma_\lambda \sim 0$). On the other hand, if the waves are much smaller than the obstructing object—for instance, an island—they are simply blocked; the only waves that continue on are those that miss the island altogether. Similarly, at sufficiently short wavelengths, the only light we detect passing through the dust cloud is the light that travels between the particles.

Combining the ideas already discussed, it is clear that the amount of extinction, as measured by A_λ , must be wavelength-dependent. Since the longer wavelengths of red light are not scattered as strongly as blue light, the starlight passing through intervening dust clouds becomes reddened as the blue light is removed. This **interstellar reddening** causes stars to appear redder than their effective temperatures would otherwise imply. Fortunately, it is possible to detect this change by carefully analyzing the absorption and emission lines in the star's spectrum.

Much of the incident blue light is scattered out of its original path and can leave the cloud in virtually any direction. As a result, looking at the cloud in a direction other than along the line of sight to a bright star behind the cloud, an observer will see a blue **reflection nebula** (recall Fig. 12.2) such as the Pleiades (Fig. 13.16b). This process is analogous to Rayleigh scattering, which produces a blue sky on Earth. The difference between Mie scattering and Rayleigh scattering is that the sizes of the scattering molecules associated with Rayleigh scattering are much smaller than the wavelength of visible light, leading to $\sigma_\lambda \propto \lambda^{-4}$.

Example 12.1.1. A certain star, located 0.8 kpc from Earth, is found to be dimmer than expected at 550 nm by $A_V = 1.1$ magnitudes, where A_V is the amount of extinction as measured through the *visual wavelength* filter (see Section 3.6). If $Q_{550} = 1.5$ and the dust grains are assumed to be spherical with radii of $0.2 \mu\text{m}$, estimate the average density ($\bar{n}$) of material between the star and Earth.

From Eq. (12.2), the optical depth along the line of sight is nearly equal to the amount of extinction in magnitudes, or $\tau_{550} \simeq 1$. Also,

$$\sigma_{550} = \pi a^2 Q_{550} \simeq 2 \times 10^{-13} \text{ m}^2.$$

Now the column density of the dust along the line of sight is given by Eq. (12.4),

$$N_d = \frac{\tau_{550}}{\sigma_{550}} \simeq 5 \times 10^{12} \text{ m}^{-2}.$$

Finally, since $N_d = \int_0^s n(s') ds' = \bar{n} \times 0.8 \text{ kpc}$, we have

$$\bar{n} = \frac{N_d}{0.8 \text{ kpc}} = 2 \times 10^{-7} \text{ m}^{-3}.$$

Number densities of this magnitude are typical of the plane of the Milky Way Galaxy.

Molecular Contributions to Interstellar Extinction Curves

Predictions of the Mie theory work well for longer wavelengths, typically from the infrared into the visible wavelength region. However, at ultraviolet wavelengths significant deviations become apparent, as can be seen by considering the ratio of A_λ , the extinction in a wavelength band centered at λ , to the extinction in some reference wavelength band, such as A_V . This ratio is often plotted versus reciprocal wavelength λ^{-1} , as in Fig. 12.3. Alternatively, **color excesses** are sometimes plotted instead, such as $(A_\lambda - A_V) / (A_B - A_V)$ or $E(B - V) \equiv (B - V)_{\text{intrinsic}} - (B - V)_{\text{observed}}$.

At longer wavelengths (the left side of the graph) the data agree well with the Mie theory. For wavelengths shorter than the blue wavelength band (B), however, the curves begin to diverge significantly, deviating from the expected relation, $A_\lambda / A_V \propto \lambda^{-1}$. Particularly evident is the “bump” in the ultraviolet at 217.5 nm or $4.6 \mu\text{m}^{-1}$. At even shorter wavelengths, the extinction curve tends to rise sharply as the wavelength decreases.

The existence of the “bump” in Fig. 12.3 gives us some hint of the composition of the dust. Graphite, a well-ordered form of carbon, interacts strongly with light near 217.5 nm.

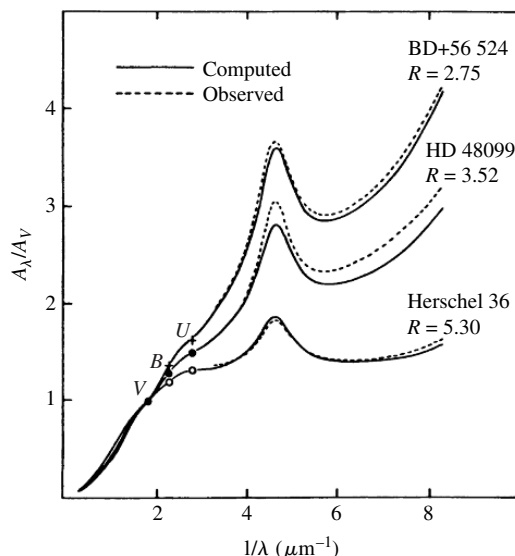


FIGURE 12.3 Interstellar extinction curves along the lines of sight to three stars. The dashed lines represent the observational data, and the solid lines are theoretical fits. The *U*, *B*, and *V* wavelength bands are indicated for reference. (Figure adapted from Mathis, *Annu. Rev. Astron. Astrophys.*, 28, 37, 1990. Reproduced with permission from the *Annual Review of Astronomy and Astrophysics*, Volume 28, ©1990 by Annual Reviews Inc.)

Although it is uncertain how carbon can organize into large graphite particles in the interstellar medium, the strength of the “bump,” the abundance of carbon, and the existence of the 217.5-nm resonance have led most researchers to suggest that graphite may be a major component of interstellar dust.

Another possible source of the 217.5-nm feature may be **polycyclic aromatic hydrocarbons** (PAHs; see Fig. 12.4). These are complex organic planar molecules with multiple benzene ring-like structures that are probably responsible for a series of molecular bands that have been observed in emission in the light from diffuse dust clouds.¹ The so-called *unidentified infrared emission bands* exist in the wavelength range between 3.3 μm and 12 μm ; they appear to be due to vibrations in the C-C and C-H bonds common in PAHs. Just as transitions between atomic energy levels are quantized, so are the energies associated with molecular bonds. In the case of molecular bonds, however, the energy levels tend to be grouped in closely spaced bands, producing characteristic broad features in the spectrum of the light. The vibration, rotation, and bending of molecular bonds are all quantized, yielding complex spectra that may be difficult to identify in large molecules.

Interstellar dust is composed of other particles as well, as evidenced by the existence of dark absorption bands at wavelengths of 9.7 μm and 18 μm in the near-infrared. These features are believed to be the result of the stretching of the Si-O molecular bond and the

¹The fact that molecules as complex as PAHs can exist in space has also been confirmed by their presence in certain types of meteorites found on Earth, known as **carbonaceous meteorites**.

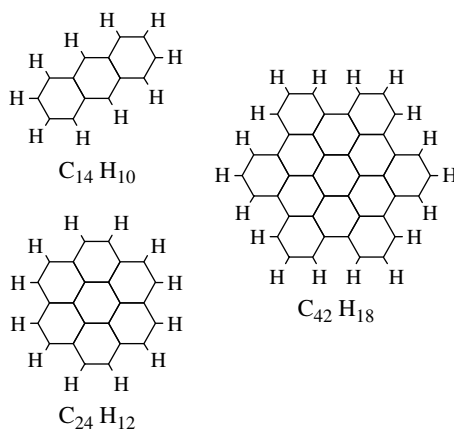


FIGURE 12.4 The structures of several polycyclic aromatic hydrocarbons: $C_{14}H_{10}$ (anthracene), $C_{24}H_{12}$ (coronene), $C_{42}H_{18}$ (hexabenzocoronene). The hexagonal structures are shorthand for indicating the presence of a carbon atom at each corner of the hexagon.

bending of Si-O-Si bonds in silicates, respectively. The existence of these absorption bands involving silicon indicates that silicate grains are also present in the dust clouds and the diffuse dust of the ISM.

An important characteristic of the light scattered from interstellar dust is that it tends to be slightly polarized. The amount of polarization is typically a few percent and depends on wavelength. This necessarily implies that the dust grains cannot be perfectly spherical. Furthermore, they must be at least somewhat aligned along a unique direction since the electric field vectors of the radiation are preferentially oriented in a particular direction. The most likely way to establish such an alignment is for the grains to interact with a weak magnetic field. Because less energy is required, the particles tend to rotate with their long axes perpendicular to the direction of the magnetic field.

All of these observations give us some clues to the nature of the dust in the ISM. Apparently the dust in the ISM is composed of both graphite and silicate grains ranging in size from several microns down to fractions of a nanometer, the characteristic size of the smaller PAHs. It appears that many of the features of the interstellar extinction curve can be reproduced by combining the contributions from all of these components.

Hydrogen as the Dominant Component of the ISM

Although dust produces most of the obscuration that is readily noticeable, the dominant component of the ISM is hydrogen gas in its various forms: neutral hydrogen (H I), ionized hydrogen (H II), and molecular hydrogen (H_2). Hydrogen comprises approximately 70% of the mass of matter in the ISM, and helium makes up most of the remaining mass; metals, such as carbon and silicon, account for only a few percent of the total.

Most hydrogen in *diffuse* interstellar hydrogen clouds is in the form of H I in the ground state. As a result, the H I is generally incapable of producing emission lines by downward transitions of electrons from one orbit to another. It is also difficult to observe H I in absorption, since UV-wavelength photons are required to lift the electrons out of the ground

state. However, in certain unique circumstances, orbiting observatories have detected absorption lines produced by cold clouds of H I when there are strong UV sources lying behind them.

21-cm Radiation of Hydrogen

Fortunately, it is still generally possible to identify neutral hydrogen in the diffuse ISM. This is done by detecting the unique radio-wavelength **21-cm line**. The 21-cm line is produced by the reversal of the spin of the electron relative to the proton in the atom's nucleus. Recall (Section 5.4) that both electrons and protons possess an inherent spin angular momentum, with the z -component of the spin angular momentum vector having one of two possible orientations, corresponding to the two allowed values of the spin quantum number, $m_s = \pm \frac{1}{2}$. Because these particles are also electrically charged, their intrinsic spins endow them with dipole magnetic fields, much like those of bar magnets. If the spins of the electron and proton are aligned (e.g., both spin axes are in the same direction), the atom has slightly more energy than if they are anti-aligned (see Fig. 12.5). As a result, if the electron's spin “flips” from being aligned with the proton to being anti-aligned, energy must be lost from the atom. If the spin flip is not due to a collision with another atom, then a photon is emitted. Of course, a photon can also be absorbed, exciting a hydrogen atom into aligning its electron and proton spins. The wavelength of the photon associated with the spin flip is 21.1 cm, corresponding to a frequency of 1420 MHz.

The emission of a 21-cm photon from an individual hydrogen atom is extremely rare. Once in the excited state, several million years can pass on average before that atom will emit a photon. Competing with this spontaneous emission are collisions between hydrogen atoms that may result in either excitation or de-excitation. In the low-density environment of the diffuse ISM, collisions occur on timescales of hundreds of years. Although this is far shorter than the spontaneous emission timescale, statistically some atoms are still able to make the necessary spontaneous transition. In contrast, the best vacuums produced in

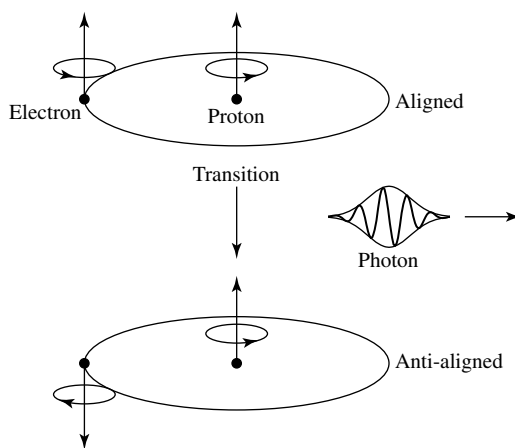


FIGURE 12.5 When the spins of the electron and proton in a hydrogen atom go from being aligned to being anti-aligned, a 21-cm-wavelength photon is emitted.

Earth-based laboratories have densities much greater than those found in the ISM, meaning that collision rates are significantly higher in laboratory environments and virtually all of the atoms in the laboratory are de-excited before they can emit 21-cm radiation. The existence of 21-cm radiation was predicted in the early 1940s and first detected in 1951. Since then it has become an important tool in mapping the location and density of H I, measuring radial velocities using the Doppler effect, and estimating magnetic fields using the Zeeman effect. 21-cm radiation is particularly valuable in determining the structure and kinematic properties of galaxies, including our own.

Although H I is quite abundant, the rarity of 21-cm emission (or absorption) from individual atoms means that the center of this line can remain optically thin over large interstellar distances. Assuming that the line profile is a Gaussian, like the shape of the Doppler line profile shown in Fig. 9.19, the optical depth of the line center is given by

$$\tau_H = 5.2 \times 10^{-23} \frac{N_H}{T \Delta v}, \quad (12.7)$$

where N_H is the column density of H I (in units of m^{-2}), T is the temperature of the gas (in kelvins), and Δv is the full width of the line at half maximum (in km s^{-1}). [Note that since the line width is due primarily to the Doppler effect (Eq. 4.30), Δv is expressed in units of velocity, rather than in wavelength units; typically $\Delta v \sim 10 \text{ km s}^{-1}$.]

As long as the 21-cm hydrogen line is optically thin (i.e., on the linear part of the curve of growth; Fig. 9.21), the optical depth is proportional to the neutral hydrogen column density. Studies of **diffuse H I clouds** indicate temperatures of 30 to 80 K, number densities in the range of $1 \times 10^8 \text{ m}^{-3}$ to $8 \times 10^8 \text{ m}^{-3}$, and masses on the order of 1–100 $M_\odot$.

Comparing τ_H with A_V along the same line of sight shows that N_H is generally proportional to N_d (the column density of dust) when $A_V < 1$. This observation suggests that dust and gas are distributed together throughout the ISM. However, when $A_V > 1$, this correlation breaks down; the column density of H I no longer increases as rapidly as the column density of dust. Apparently, other physical processes are involved when the dust becomes optically thick.

Optically thick dust clouds shield hydrogen from sources of ultraviolet radiation. One consequence of this shielding is that molecular hydrogen can exist without the threat of undergoing dissociation by UV photon absorption. Dust can also enhance the H_2 formation rate beyond what would be expected by random collisions of hydrogen atoms. This enhancement occurs for two reasons: (1) A dust grain can provide a *site* on the surface of the grain where the hydrogen atoms can meet, rather than requiring chance encounters in the ISM, and (2) the dust provides a *sink* for the binding energy that must be liberated if a stable molecule is to form. The liberated energy goes into heating the grain and ejecting the H_2 molecule from the formation site. If the column density of atomic hydrogen is sufficiently large (N_H on the order of 10^{25} m^{-2}), it can also shield H_2 from UV photodissociation. Consequently, **molecular clouds** are surrounded by shells of H I.

Molecular Tracers of H_2

Since the structure of H_2 differs greatly from that of atomic hydrogen, the H_2 molecule does not emit 21-cm radiation. This explains why N_H and A_V are poorly correlated in molecular

clouds when $A_V > 1$; the number density of atomic hydrogen decreases significantly as the hydrogen becomes locked up in its molecular form.

Unfortunately, H_2 is very difficult to observe directly because the molecule does not have any emission or absorption lines in the visible or radio portions of the electromagnetic spectrum at the cool temperatures typical of the ISM. In special circumstances when $T > 2000$ K, it is possible to detect rotational and vibrational bands (known collectively as *rovibrational bands*) associated with the molecular bond. However, in most instances it becomes necessary to use other molecules as *tracers* of H_2 by making the assumption that their abundances are proportional to the abundance of H_2 . Because of its relatively high abundance (approximately 10^{-4} that of H_2), the most commonly investigated tracer is carbon monoxide, CO, although other molecules have also been used, including CH, OH, CS, C_3H_2 , HCO^+ , and N_2H^+ . It is also possible to use **isotopomers** of the molecules, such as ^{13}CO or $C^{18}O$, to further refine studies of molecular clouds. Given that molecules have moments of inertia that affect their spectra, different isotopes in molecules result in different spectral wavelengths; see, for example, Problem 12.7. (Note that when the specific isotope is not indicated, it is assumed that the most abundant isotope is implied; thus CO implies $^{12}C^{16}O$.)

During collisions the tracer molecules become excited (or de-excited) and spontaneous transitions from excited states result in the emission of photons in wavelength regions that are more easily observed than those associated with H_2 , such as the 2.6-mm transition of CO. Since collision rates depend on both the gas temperature (or thermal kinetic energy) and the number densities of the species, molecular tracers can provide information about the environment within a molecular cloud. In fact, an estimate of atomic and molecular collision rates can be made in a way completely analogous to the approach used to obtain the nuclear reaction rate equation (Eq. 10.29).

The Classification of Interstellar Clouds

The results of these studies show that conditions within molecular clouds can vary widely. Consequently, any effort to specify a discrete classification scheme is destined to fail because the delineation between types is blurred at best. However, even with that caveat, a broad classification scheme is still useful for distinguishing the general characteristics of specific environments.

In clouds where the hydrogen gas is primarily atomic and the interstellar extinction is roughly $1 < A_V < 5$, molecular hydrogen may be found in regions of higher column density. Such clouds are sometimes referred to as **diffuse molecular clouds**, or alternatively as **translucent molecular clouds**. Conditions in diffuse molecular clouds are typical of diffuse H I clouds but with somewhat higher masses; they have temperatures of 15 to 50 K, $n \sim 5 \times 10^8$ to 5×10^9 m^{-3} , $M \sim 3$ to $100 M_\odot$, and they measure several parsecs across. Both H I clouds and diffuse molecular clouds tend to be irregularly shaped.

Giant molecular clouds (GMCs) are enormous complexes of dust and gas where temperatures are typically $T \sim 15$ K, number densities are in the range $n \sim 1 \times 10^8$ to 3×10^8 m^{-3} , masses are typically $10^5 M_\odot$ but may reach $10^6 M_\odot$, and typical sizes are on the order of 50 pc across. The famous Horsehead Nebula, also known as Barnard 33 (B33),



FIGURE 12.6 The Horsehead Nebula is part of the Orion giant molecular cloud complex. The “horsehead” appearance is due to dust protruding into an H II (ionized hydrogen) environment. (European Southern Observatory)

is shown in Fig. 12.6. The Horsehead Nebula is a portion of the Orion giant molecular cloud complex. Thousands of GMCs are known to exist in our Galaxy, mostly in its spiral arms.

Overall, the structure of GMCs tend to be clumpy with local regions of significantly greater density. **Dark cloud complexes** of roughly $10^4 M_{\odot}$ have $A_V \sim 5$, $n \sim 5 \times 10^8 \text{ m}^{-3}$, diameters on the order of 10 pc, and characteristic temperatures of 10 K. Smaller, individual **clumps** may be even more dense, with $A_V \sim 10$, $n \sim 10^9 \text{ m}^{-3}$, diameters of a couple of parsecs, temperatures of 10 K or so, and masses of $30 M_{\odot}$. At even smaller scales are **dense cores** with masses on the order of $10 M_{\odot}$, $A_V > 10$, $n \sim 10^{10} \text{ m}^{-3}$, characteristic diameters of 0.1 pc, and temperatures of 10 K. Finally, in some localized regions of GMCs, observations reveal **hot cores** with characteristic sizes of 0.05 to 0.1 pc, where $A_V \sim 50$ to 1000, $T \sim 100$ to 300 K, $n \sim 10^{13}$ to 10^{15} m^{-3} , and $M \sim 10$ to $3000 M_{\odot}$. Based on observations from infrared telescopes such as NASA’s Spitzer Space Telescope and the European Space Agency’s Infrared Space Observatory, hot cores appear to have massive, young O and B stars embedded within them, suggesting strongly that these are regions of recent star formation.

Located outside of larger molecular complexes are the almost spherical clouds known as **Bok globules** (see, for example, Fig. 12.7).² These globules are characterized by large visual extinctions ($A_V \sim 10$), low temperatures ($T \sim 10$ K), relatively large number densities ($n > 10^{10} \text{ m}^{-3}$), low masses ($M \sim 1$ to $1000 M_{\odot}$), and small sizes of typically less than 1 pc. Infrared surveys of Bok globules have revealed that many, perhaps most, of these objects harbor young low-luminosity stars in their centers, implying that Bok globules are also sites of active star formation. In fact, Bok globules appear to be dense cores that have

²Bok globules are named after Bart Bok (1906–1983), who first studied these objects in the 1940s.

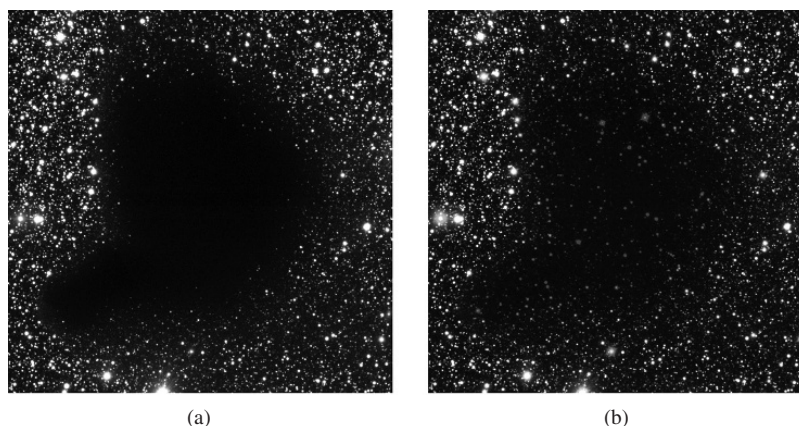


FIGURE 12.7 The Bok globule, Barnard 68 (B68), observed in visible light [(a) composite of BVI bands] and in infrared light [(b) composite of BIK bands]. The visible image was obtained by one of the 8-m telescopes of the European Southern Observatory's Very Large Telescope at Paranal. The infrared image was obtained by ESO's 3.58-m New Technology Telescope at La Silla. Notice that significantly reddened stars can be seen through the globule in the infrared. (Interstellar reddening is the result of scattering photons off of dust grains.) (European Southern Observatory)

been stripped of their surrounding molecular gas by nearby hot, massive stars. The process by which stars form out of the ISM will be considered in the next section.

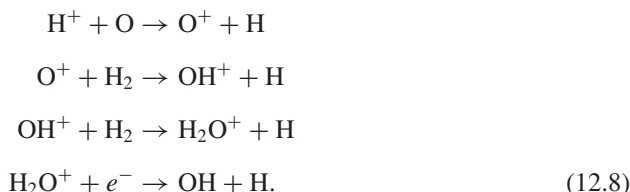
Interstellar Chemistry

Along with the molecules and dust grains already discussed, the ISM is rich in other molecules as well. As of June 2005, radio observations have resulted in the positive identification of 125 molecules (not including isotopomers), ranging in complexity from diatomic molecules such as H_2 and CO , and triatomic molecules such as H_2O and H_3^+ , to fairly long organic strings, including HC_{11}N .

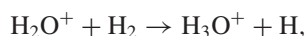
Given the complex nature of the molecules present in the interstellar medium, it is evident that the chemistry of the ISM is also quite complex. The specific processes in operation in a given molecular cloud depend on the density and temperature of the gas, as well as its composition and the presence of dust grains. We noted on page 406 that dust grains must be present for the formation of molecular hydrogen, H_2 , the dominant constituent in molecular clouds. It is also likely that dust grains can help facilitate the formation of numerous other molecules as well, including CH , NH , OH , CH_2 , CO , CO_2 , and H_2O . In fact, in sufficiently dense clouds, the formation of molecules on the surfaces of grains can actually lead to the development of icy mantles on the grains. Absorption signatures of solid CO , CO_2 , H_2O , CH_4 , CH_3OH , NH_3 , and other ices have been measured in combination with the infrared spectra of silicate dust grains.

In addition to the chemistry that can occur on grain surfaces, it is also possible for molecules to form in the gas phase. For example, the hydroxyl molecule (OH) can form through a series of reactions involving atomic and molecular ions, including the ionic water

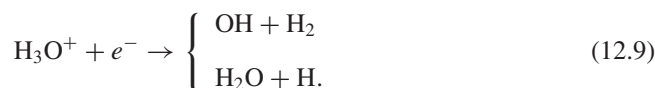
molecule, H_2O^+ :



Eq. (12.8) competes with another reaction involving molecular hydrogen,



leading to the production of either a hydroxyl molecule (75% of the time) or a water molecule via

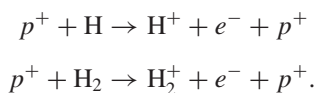


The Heating and Cooling of the ISM

Not only are molecules and dust grains critical in understanding the chemistry of the ISM, but they also play important roles in the heating and cooling of the material between the stars. You may have noticed that diffuse molecular clouds have higher gas temperatures than giant molecular clouds, and the dense cores of GMCs are even cooler yet. On the other hand, the hot cores of GMCs have significantly greater temperatures. What are the physical causes of these observational trends?

Much of the heating of the interstellar medium comes from **cosmic rays**, charged particles that travel through space with sometimes astonishing amounts of energy. A single proton may possess an energy ranging anywhere from 10 to 10^{14} MeV.³ The highest energy cosmic rays are extremely rare, but energies in the range 10^3 to 10^8 MeV are common. The sources of cosmic rays include stellar flares and supernova explosions and will be discussed in Section 15.5.

Heating by cosmic rays comes primarily through the ionization of hydrogen atoms and molecules as a result of collisions with cosmic ray protons;



When an atom or molecule is ionized, an electron is ejected that carries some of the original kinetic energy of the proton with it. It is this ejected electron that interacts with the ISM to increase the average kinetic energy of the ISM's constituents via collisions with molecules (see, for example, Eqs. 12.8 and 12.9). Those molecules then collide with other molecules

³ 10^{14} MeV is roughly the kinetic energy of a tennis ball of mass 0.057 kg traveling at 100 km h^{-1} (approximately 60 mph).

in the gas, distributing thermal kinetic energy throughout the cloud, thereby raising the temperature of the cloud.

Other sources of heating in molecular clouds include the ionization of carbon atoms by ultraviolet starlight resulting in ejected electrons, the photoelectric ejection of electrons from dust grains by ultraviolet starlight, the absorption of light energy into the lattice of dust grains, and the ionization of hydrogen by stellar X-rays. Shocks from supernovae or strong stellar winds can also produce some heating of molecular clouds in special cases.

To balance the heating processes, cooling mechanisms must also be in operation. The primary mechanism for cooling is based on the emission of infrared photons. Recalling Mie scattering (Eq. 12.5), when photon wavelengths are on the order of, or longer than, the size of dust grains, they are less likely to be scattered. IR photons can pass more easily through the molecular cloud than can shorter-wavelength photons, allowing the IR photons to transport energy out of the cloud.

IR photons are produced in molecular clouds through collisions between ions, atoms, molecules, and dust grains. Typically a collision between ions, atoms, or molecules results in one of the species being left in an excited state; the energy of the excited state comes from the kinetic energy of the collision. The species in the excited state then decays back to the ground state through the emission of an IR photon. For example,



Here O^* represents an excited state of the oxygen atom. The collisional kinetic energy (thermal energy) is thus transformed into an IR photon that escapes the cloud. Collisional excitations of C^+ and CO by H and H_2 , respectively, are also significant contributors to cooling of molecular clouds.

Collisions involving dust grains can also result in cooling of molecular clouds. This process is similar to ionic, atomic, and molecular collisions in that the lattice of a dust grain can be left with excess thermal energy after the collision. The grain then emits infrared energy that is able to escape from the cloud.

The Sources of Dust Grains

It is apparent that even though dust grains make up only about one percent of the mass of a molecular cloud, they are important constituents in determining its chemistry and physics. The question of the source of these grains then naturally arises. Although observations indicate that dust grains can be formed in the envelopes of very cool stars, aided by the enhanced density in those environments relative to molecular clouds, grains can also be easily destroyed by UV and X-ray photons. Dust grains are also formed as a product of supernova explosions and stellar winds. However, none of these sources appear to be able to provide the abundance of massive grains found in molecular clouds. Rather, it appears that grains probably grow by a process of coagulation within the molecular clouds themselves. Dust grain formation represents just one of many areas of active research into the nature of the ISM.

12.2 ■ THE FORMATION OF PROTOSTARS

Our understanding of stellar evolution has developed significantly since the 1960s, reaching the point where much of the life history of a star is well determined. This success has been due to advances in observational techniques, improvements in our knowledge of the physical processes important in stars, and increases in computational power. In the remainder of this chapter and in Chapter 13 we will present an overview of the lives of stars, leaving detailed discussions of some special phases of evolution until later, specifically stellar pulsation, supernovae, and compact objects (stellar corpses).

The Jeans Criterion

Despite many successes, important questions remain concerning how stars change during their lifetimes. One area where the picture is far from complete is in the earliest stage of evolution, the formation of pre-nuclear-burning objects known as **protostars** from interstellar molecular clouds.

If globules and cores in molecular clouds are the sites of star formation, what conditions must exist for collapse to occur? Sir James Jeans (1877–1946) first investigated this problem in 1902 by considering the effects of small deviations from hydrostatic equilibrium. Although several simplifying assumptions are made in the analysis, such as neglecting effects due to rotation, turbulence, and galactic magnetic fields, it provides important insights into the development of protostars.

The virial theorem (Eq. 2.46),

$$2K + U = 0,$$

describes the condition of equilibrium for a stable, gravitationally bound system.⁴ We have already seen that the virial theorem arises naturally in the discussion of orbital motion, and we have also invoked it in estimating the amount of gravitational energy contained within a star (Eq. 10.22). The virial theorem may also be used to estimate the conditions necessary for protostellar collapse.

If twice the total internal kinetic energy of a molecular cloud ($2K$) exceeds the absolute value of the gravitational potential energy ($|U|$), the force due to the gas pressure will dominate the force of gravity and the cloud will expand. On the other hand, if the internal kinetic energy is too low, the cloud will collapse. The boundary between these two cases describes the critical condition for stability when rotation, turbulence, and magnetic fields are neglected.

Assuming a spherical cloud of constant density, the gravitational potential energy is approximately (Eq. 10.22)

$$U \sim -\frac{3}{5} \frac{GM_c^2}{R_c},$$

where M_c and R_c are the mass and radius of the cloud, respectively. We may also estimate

⁴We have implicitly assumed that the kinetic and potential energy terms are averaged over time, as in Section 2.4.

the cloud's internal kinetic energy, given by

$$K = \frac{3}{2} N k T,$$

where N is the total number of particles. But N is just

$$N = \frac{M_c}{\mu m_H},$$

where μ is the mean molecular weight. Now, by the virial theorem, the condition for collapse ($2K < |U|$) becomes

$$\frac{3 M_c k T}{\mu m_H} < \frac{3}{5} \frac{G M_c^2}{R_c}. \quad (12.12)$$

The radius may be replaced by using the initial mass density of the cloud, ρ_0 , assumed here to be constant throughout the cloud,

$$R_c = \left(\frac{3 M_c}{4 \pi \rho_0} \right)^{1/3}. \quad (12.13)$$

After substitution into Eq. (12.12), we may solve for the minimum mass necessary to initiate the spontaneous collapse of the cloud. This condition is known as the **Jeans criterion**:

$$M_c > M_J,$$

where

$$M_J \simeq \left(\frac{5 k T}{G \mu m_H} \right)^{3/2} \left(\frac{3}{4 \pi \rho_0} \right)^{1/2} \quad (12.14)$$

is called the **Jeans mass**. Using Eq. (12.13), the Jeans criterion may also be expressed in terms of the minimum radius necessary to collapse a cloud of density ρ_0 :

$$R_c > R_J, \quad (12.15)$$

where

$$R_J \simeq \left(\frac{15 k T}{4 \pi G \mu m_H \rho_0} \right)^{1/2} \quad (12.16)$$

is the **Jeans length**.

The Jeans mass derivation given above neglected the important fact that there must exist an external pressure on the cloud due to the surrounding interstellar medium (such as the encompassing GMC in the case of an embedded dense core). Although we will not derive

the expression here, the critical mass required for gravitational collapse in the presence of an external gas pressure of P_0 is given by the **Bonnor–Ebert mass**,

$$M_{\text{BE}} = \frac{c_{\text{BE}} v_T^4}{P_0^{1/2} G^{3/2}}, \quad (12.17)$$

where

$$v_T \equiv \sqrt{kT/\mu m_H} \quad (12.18)$$

is the *isothermal sound speed* ($\gamma = 1$ in Eq. 10.84), and the dimensionless constant c_{BE} is given by

$$c_{\text{BE}} \simeq 1.18.$$

It is shown in Problem 12.11 that the Jeans mass (Eq. 12.14) can be written in the form of Eq. (12.17) with $c_J \simeq 5.46$ replacing c_{BE} . The smaller constant for the Bonnor–Ebert mass is to be expected since an external compression force due to P_0 is being exerted on the cloud.⁵

Example 12.2.1. For a typical diffuse hydrogen cloud, $T = 50$ K and $n = 5 \times 10^8 \text{ m}^{-3}$. If we assume that the cloud is entirely composed of H I, $\rho_0 = m_H n_H = 8.4 \times 10^{-19} \text{ kg m}^{-3}$. Taking $\mu = 1$ and using Eq. (12.14), the minimum mass necessary to cause the cloud to collapse spontaneously is approximately $M_J \sim 1500 M_\odot$. However, this value significantly exceeds the estimated 1 to 100 $M_\odot$ believed to be contained in H I clouds. Hence diffuse hydrogen clouds are stable against gravitational collapse.

On the other hand, for a dense core of a giant molecular cloud, typical temperatures and number densities are $T = 10$ K and $n_{\text{H}_2} = 10^{10} \text{ m}^{-3}$. Since dense clouds are predominantly molecular hydrogen, $\rho_0 = 2m_H n_{\text{H}_2} = 3 \times 10^{-17} \text{ kg m}^{-3}$ and $\mu \simeq 2$. In this case the Jeans mass is $M_J \sim 8 M_\odot$, characteristic of the masses of dense cores being on the order of 10 $M_\odot$. Apparently the dense cores of GMCs are unstable to gravitational collapse, consistent with being sites of star formation.

If the Bonnor–Ebert mass (Eq. 12.17) is used as the critical collapse condition, then the required mass reduces to approximately 2 $M_\odot$.

Homologous Collapse

In the case that the criterion for gravitational collapse has been satisfied in the absence of rotation, turbulence, or magnetic fields, the molecular cloud will collapse. If we make the simplifying (and possibly unrealistic) assumption that any existing pressure gradients are too small to influence the motion appreciably, then the cloud is essentially in free-fall during the first part of its evolution. Furthermore, throughout the free-fall phase, the temperature

⁵You may be interested to know that the derivation of Eq. (12.17) involves the *isothermal Lane–Emden equation*, analogous to Eq. (10.110) used in deriving polytropes in Chapter 10.

of the gas remains nearly constant (i.e., the collapse is said to be *isothermal*). This is true as long as the cloud remains optically thin and the gravitational potential energy released during the collapse can be efficiently radiated away. In this case the spherically symmetric hydrodynamic equation (Eq. 10.5) can be used to describe the contraction if we assume that $|dP/dr| \ll GM_r\rho/r^2$. After canceling the density on both sides of the expression, we have

$$\frac{d^2r}{dt^2} = -G \frac{M_r}{r^2}. \quad (12.19)$$

Of course, the right-hand side of Eq. (12.19) is just the local acceleration of gravity at a distance r from the center of a spherical cloud. As usual, the mass of the sphere interior to the radius r is denoted by M_r .

To describe the behavior of the surface of a sphere of radius r within the collapsing cloud as a function of time, Eq. (12.19) must be integrated over time. Since we are interested only in the surface that encloses M_r , the mass interior to r will remain a constant during that collapse. As a result, we may replace M_r by the product of the initial density ρ_0 and the initial spherical volume, $4\pi r_0^3/3$. Then, if we multiply both sides of Eq. (12.19) by the velocity of the surface of the sphere, we arrive at the expression

$$\frac{dr}{dt} \frac{d^2r}{dt^2} = - \left(\frac{4\pi}{3} G \rho_0 r_0^3 \right) \frac{1}{r^2} \frac{dr}{dt},$$

which can be integrated once with respect to time to give

$$\frac{1}{2} \left(\frac{dr}{dt} \right)^2 = \left(\frac{4\pi}{3} G \rho_0 r_0^3 \right) \frac{1}{r} + C_1.$$

The integration constant, C_1 , can be evaluated by requiring that the velocity of the sphere's surface be zero at the beginning of the collapse, or $dr/dt = 0$ when $r = r_0$. This gives

$$C_1 = -\frac{4\pi}{3} G \rho_0 r_0^2.$$

Substituting and solving for the velocity at the surface, we have

$$\frac{dr}{dt} = - \left[\frac{8\pi}{3} G \rho_0 r_0^2 \left(\frac{r_0}{r} - 1 \right) \right]^{1/2}. \quad (12.20)$$

Note that the negative root was chosen because the cloud is collapsing.

To integrate Eq. (12.20) so that we can obtain an expression for the position as a function of time, we make the substitutions

$$\theta \equiv \frac{r}{r_0}$$

and

$$\chi \equiv \left(\frac{8\pi}{3} G \rho_0 \right)^{1/2},$$

which leads to the differential equation

$$\frac{d\theta}{dt} = -\chi \left(\frac{1}{\theta} - 1 \right)^{1/2}. \quad (12.21)$$

Making yet another substitution,

$$\theta \equiv \cos^2 \xi, \quad (12.22)$$

and after some manipulation, Eq. (12.21) becomes

$$\cos^2 \xi \frac{d\xi}{dt} = \frac{\chi}{2}. \quad (12.23)$$

Equation (12.23) may now be integrated directly with respect to t to yield

$$\frac{\xi}{2} + \frac{1}{4} \sin 2\xi = \frac{\chi}{2} t + C_2. \quad (12.24)$$

Lastly, the integration constant, C_2 , must be evaluated. Doing so requires that $r = r_0$ when $t = 0$, which implies that $\theta = 1$, or $\xi = 0$ at the beginning of the collapse. Therefore, $C_2 = 0$.

We have finally arrived at the equation of motion for the gravitational collapse of the cloud, given in parameterized form by

$$\xi + \frac{1}{2} \sin 2\xi = \chi t. \quad (12.25)$$

Our task now is to extract the behavior of the collapsing cloud from this equation. From Eq. (12.25), it is possible to calculate the **free-fall timescale** for a cloud that has satisfied the Jeans criterion. Let $t = t_{\text{ff}}$ when the radius of the collapsing sphere reaches zero ($\theta = 0$, $\xi = \pi/2$).⁶ Then

$$t_{\text{ff}} = \frac{\pi}{2\chi}.$$

Substituting the value for χ , we have

$$t_{\text{ff}} = \left(\frac{3\pi}{32} \frac{1}{G\rho_0} \right)^{1/2}. \quad (12.26)$$

You should notice that the free-fall time is actually independent of the initial radius of the sphere. Consequently, as long as the original density of the spherical molecular cloud was uniform, all parts of the cloud will take the same amount of time to collapse, and the density will increase at the same rate everywhere. This behavior is known as a **homologous collapse**.

⁶This is obviously an unphysical final condition, since it implies infinite density. If $r_0 \gg r_{\text{final}}$, however, then $r_{\text{final}} \simeq 0$ is a reasonable approximation for our purposes here.

However, if the cloud is somewhat centrally condensed when the collapse begins, the free-fall time will be shorter for material near the center than for material farther out. Thus, as the collapse progresses, the density will increase more rapidly near the center than in other regions. In this case the collapse is referred to as an **inside-out collapse**.

Example 12.2.2. Using data given in Example 12.2.1 for a dense core of a giant molecular cloud, we may estimate the amount of time required for the collapse. Assuming a density of $\rho_0 = 3 \times 10^{-17} \text{ kg m}^{-3}$ that is constant throughout the core, Eq. (12.26) gives

$$t_{\text{ff}} = 3.8 \times 10^5 \text{ yr.}$$

To investigate the actual behavior of the collapse in our simplified model, we must first solve Eq. (12.25) for ξ , given a value for t , and then use Eq. (12.22) to find $\theta = r/r_0$. However, Eq. (12.25) cannot be solved explicitly, so numerical techniques must be employed. The numerical solution of the homologous collapse of the molecular cloud is shown in Fig. 12.8. Notice that the collapse is quite slow initially and accelerates quickly as t_{ff} is approached. At the same time, the density increases very rapidly during the final stages of collapse.

The Fragmentation of Collapsing Clouds

Since the masses of fairly large molecular clouds could exceed the Jeans limit, from Eq. (12.14) our simple analysis seems to imply that stars can form with very large masses, possibly up to the initial mass of the cloud. However, observations show that this does not happen. Furthermore, it appears that stars frequently (perhaps even preferentially) tend to form in groups, ranging from binary star systems to clusters that contain hundreds of thousands of members (see Section 13.3).

The process of **fragmentation** that segments a collapsing cloud is an aspect of star formation that is under significant investigation. To see that fragmentation must occur by

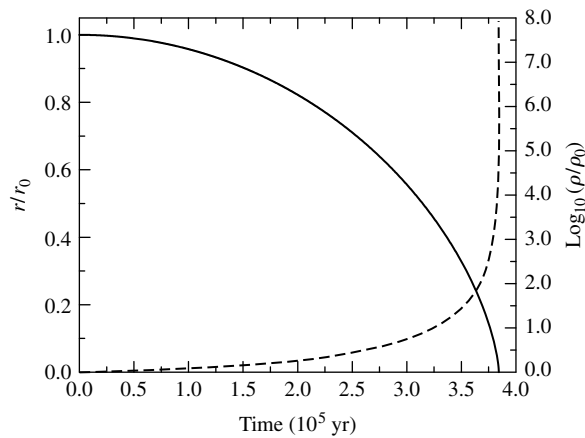


FIGURE 12.8 The homologous collapse of a molecular cloud, as discussed in Example 12.2.2. r/r_0 is shown as the solid line and $\log_{10}(\rho/\rho_0)$ is shown as the dashed line. The initial density of the cloud was $\rho_0 = 3 \times 10^{-17} \text{ kg m}^{-3}$ and the free-fall time is $3.8 \times 10^5 \text{ yr}$.

some mechanism(s), refer again to the equation for the Jeans mass (Eq. 12.14). An important consequence of the collapse of a molecular cloud is that the density of the cloud increases by many orders of magnitude during free-fall (Fig. 12.8). Consequently, since T remains nearly constant throughout much of the collapse, it appears that the Jeans mass must decrease. After collapse has begun, any initial inhomogeneities in density will cause individual sections of the cloud to satisfy the Jeans mass limit independently and begin to collapse locally, producing smaller features within the original cloud. This cascading collapse could lead to the formation of large numbers of smaller objects.

It is important to point out that one challenge with the overly simplified scenario described here is that the process implies that far too many stars would be produced. It is likely that only about 1% of the cloud actually forms stars.

What is it that stops the fragmentation process? Since we observe a galaxy filled with stars that have masses on the order of the mass of the Sun, the cascading fragmentation of the cloud cannot proceed without interruption. The answer to the question lies in our implicit assumption that the collapse is isothermal, which in turn implies that the only term that changes in Eq. (12.14) is the density. Clearly this cannot be the case since stars have temperatures much higher than 10 to 100 K. If the energy that is released during a gravitational collapse is radiated away efficiently, the temperature can remain nearly constant. At the other extreme, if the energy cannot be transported out of the cloud at all (an *adiabatic* collapse), then the temperature must rise. Of course, the real situation must be somewhere between these two limits, but by considering each of these special cases carefully, we can begin to understand some of the important features of the problem.

If the collapse changes from being essentially isothermal to adiabatic, the associated temperature rise would begin to affect the value of the Jeans mass. In Chapter 10 we saw that for an adiabatic process the pressure of the gas is related to its density by γ , the ratio of specific heats (Eq. 10.86). Using the ideal gas law (Eq. 10.11), an adiabatic relation between density and temperature can be obtained,

$$T = K'' \rho^{\gamma-1}, \quad (12.27)$$

where K'' is a constant. Substituting this expression into Eq. (12.14), we find that for an adiabatic collapse, the dependence of the Jeans mass on density becomes

$$M_J \propto \rho^{(3\gamma-4)/2}.$$

For atomic hydrogen $\gamma = 5/3$, giving $M_J \propto \rho^{1/2}$; the Jeans mass *increases* with increasing density for a perfectly adiabatic collapse of a cloud. This behavior means that the collapse results in a minimum value for the mass of the fragments produced. The minimum mass depends on the point when the collapse goes from being predominantly isothermal to adiabatic.

Of course, this transition is not instantaneous or even complete. However, it is possible to make a crude order-of-magnitude estimate of the lower mass limit of the fragments. As we have already mentioned, according to the virial theorem, energy must be liberated during the collapse of the cloud. From Eq. (10.22) and the discussion contained in Example 10.3.1, the energy released is roughly

$$\Delta E_g \simeq \frac{3}{10} \frac{GM_J^2}{R_J}$$

for a spherical cloud just satisfying the Jeans criterion at some point during the collapse. Averaged over the free-fall time, the luminosity due to gravity is given by

$$L_{\text{ff}} \simeq \frac{\Delta E_g}{t_{\text{ff}}} \sim G^{3/2} \left(\frac{M_J}{R_J} \right)^{5/2},$$

where we have made use of Eq. (12.26) and have neglected terms of order unity.

If the cloud were optically thick and in thermodynamic equilibrium, the energy would be emitted as blackbody radiation. However, during collapse the process of releasing the energy is less efficient than for an ideal blackbody. Following Eq. (3.17), we may express the radiated luminosity as

$$L_{\text{rad}} = 4\pi R^2 e \sigma T^4,$$

where an efficiency factor, $0 < e < 1$, has been introduced to indicate the deviation from thermodynamic equilibrium. If the collapse is perfectly isothermal and escaping radiation does not interact at all with overlying infalling material, $e \sim 0$. If, on the other hand, energy emitted by some parts of the cloud is absorbed and then re-emitted by other parts of the cloud, thermodynamic equilibrium would more nearly apply and e would be closer to unity.

Equating the two expressions for the cloud's luminosity,

$$L_{\text{ff}} = L_{\text{rad}},$$

and rearranging, we have

$$M_J^{5/2} = \frac{4\pi}{G^{3/2}} R_J^{9/2} e \sigma T^4.$$

Making use of Eq. (12.13) to eliminate the radius, and then using Eq. (12.14) to write the density in terms of the Jeans mass, we arrive at an estimate of when adiabatic effects become important, expressed in terms of the minimum obtainable Jeans mass:

$$M_{J_{\text{min}}} = 0.03 \left(\frac{T^{1/4}}{e^{1/2} \mu^{9/4}} \right) M_{\odot}, \quad (12.28)$$

where T is expressed in kelvins. If we take $\mu \sim 1$, $e \sim 0.1$, and $T \sim 1000$ K at the time when adiabatic effects may start to become significant, $M_J \sim 0.5 M_{\odot}$; fragmentation ceases when the segments of the original cloud begin to reach the range of solar mass objects. The estimate is relatively insensitive to other reasonable choices for T , e , and μ . For instance, if $e \sim 1$ then $M_J \sim 0.2 M_{\odot}$.

Additional Physical Processes in Protostellar Star Formation

We have, of course, left out a number of important features in our calculations. For instance, we have freely used the Jeans criterion during each point in the collapse of the cloud to discuss the process of fragmentation. This cannot be correct, since our estimate of the Jeans criterion was based on a perturbation of a static cloud; no consideration was made of the initial velocity of the cloud's outer layers. We have also neglected the details

of radiation transport through the cloud, as well as vaporization of the dust grains, dissociation of molecules, and ionization of the atoms. Nevertheless, it is worth noting that as unsophisticated as the preceding analysis was, it did illustrate important aspects of the fundamental problem and left us with a result that is reasonable. Such preliminary approaches to understanding complex physical systems are powerful tools in our study of nature.⁷ More sophisticated estimates of the complex process of cessation of fragmentation place the limit an order of magnitude lower than determined above, at about $0.01 M_{\odot}$.

Perhaps just as important to the problem of the collapse process are the possible effects of rotation (angular momentum), the deviation from spherical symmetry, turbulent motions in the gas, and the presence of magnetic fields. For example, an appreciable amount of angular momentum present in the original cloud is likely to result in a disk-like structure for at least a part of the original material, since collapse will proceed at a more rapid rate along the axis of rotation relative to collapse along the equator (aspects of the angular momentum of the collapsing cloud will be explored in Problem 12.18).

It is also apparent from careful investigations of molecular clouds that magnetic fields must also play a crucial role and, in fact, are likely to control the onset of collapse. That mechanisms other than gravity must be involved becomes clear in simply considering the free-fall time of the dense core discussed in Example 12.2.2. From that calculation, the collapse of the dense core should occur on a timescale on the order of 10^5 yr. While this may seem long by human standards, it is quite short on stellar evolution timescales. This would imply that almost as soon as a dense core forms, it begins producing stars. This would also imply that dense cores should be very rare; however, many dense cores are observable throughout our Galaxy.

Zeeman measurements of various molecular clouds indicate the presence of magnetic fields with strengths typically on the order of magnitude of 1 to 100 nT. If the magnetic field of a cloud is “frozen in,” and the cloud is compressed, the magnetic field strength will increase, leading to an increase in the magnetic pressure and resistance to the compression. In fact, if the cloud is stable to collapse because of magnetic pressure, it will remain so as long as the magnetic field does not decay (recall the discussion of magnetic fields associated with sunspots in Chapter 11).

During the derivation of the Jeans criterion, the virial theorem was invoked using a balance between gravitational potential energy and the cloud’s internal (thermal) kinetic energy. Absent from that calculation was the inclusion of energy due to the presence of magnetic fields. When magnetic fields are included, the critical mass can be expressed as

$$M_B = c_B \frac{\pi R^2 B}{G^{1/2}}, \quad (12.29)$$

where $c_B = 380 \text{ N}^{1/2} \text{ m}^{-1} \text{ T}^{-1}$ for a magnetic field permeating a spherical, uniform cloud. If B is expressed in nT and R in units of pc, then Eq. (12.29) can be written in the more illustrative form

$$M_B \simeq 70 M_{\odot} \left(\frac{B}{1 \text{ nT}} \right) \left(\frac{R}{1 \text{ pc}} \right)^2. \quad (12.30)$$

⁷This type of approach is sometimes called a “back-of-the-envelope” calculation because of the relatively small space required to carry out the estimate. Extensive use of “back-of-the-envelope” calculations is made throughout this text to illustrate the effects of key physical processes.

If the mass of the cloud is less than M_B , the cloud is said to be **magnetically subcritical** and stable against collapse, but if the mass of the cloud exceeds M_B , the cloud is **magnetically supercritical** and the force due to gravity will overwhelm the ability of the magnetic field to resist collapse.

Example 12.2.3. For the dense core considered in Examples 12.2.1 and 12.2.2, if the dense core has a magnetic field of 100 nT threading through it, and if it has a radius of 0.1 pc, the magnetic critical mass would be $M_B \simeq 70 M_\odot$, implying that a dense core of mass $10 M_\odot$ would be stable against collapse. However, if $B = 1$ nT, then $M_B \simeq 0.7 M_\odot$ and collapse would occur.

Ambipolar Diffusion

The last example hints at another possibility for triggering the collapse of a dense core. If a core that was originally subcritical were to become supercritical, collapse could ensue. This could happen in one of two ways: a group of subcritical clouds could combine to form a supercritical cloud, or the magnetic field could be rearranged so that the field strength is lessened in a portion of the cloud. It appears that both processes may occur, although the latter process seems to dominate the pre-collapse evolution of most molecular clouds.

Recall that only charged particles such as electrons or ions are tied to magnetic field lines; neutrals are not affected directly. Given that dense molecular cores are dominated by neutrals, how can magnetic fields have any substantial effect on the collapse? The answer lies in the collisions between neutrals and the ions (electrons do not significantly affect neutral atoms or molecules through collisions). As neutrals try to drift across magnetic field lines, they collide with the “frozen-in” ions, and the motions of the neutrals are inhibited. However, if there is a net defined direction for the motion of neutrals due to gravitational forces, they will still tend to migrate slowly in that direction. This slow migration process is known as **ambipolar diffusion**.

To determine the relative impact of ambipolar diffusion, we need to estimate a characteristic timescale for the diffusion process. This is done by comparing the size of the molecular cloud to the time it takes for a neutral to drift across the cloud. It can be shown that the timescale for ambipolar diffusion is approximately

$$t_{\text{AD}} \simeq \frac{2R}{v_{\text{drift}}} \simeq 10 \text{ Gyr} \left(\frac{n_{\text{H}_2}}{10^{10} \text{ m}^{-3}} \right) \left(\frac{B}{1 \text{ nT}} \right)^{-2} \left(\frac{R}{1 \text{ pc}} \right)^2. \quad (12.31)$$

Once collapse begins, magnetic fields can be further altered by undergoing reconnection events similar to those of solar flares.

Example 12.2.4. Returning to the dense core we used in previous examples, if $B = 1$ nT and $R = 0.1$ pc, we find from Eq. (12.31) that the timescale for ambipolar diffusion is 100 Myr. This is several hundred times longer than the free-fall timescale determined in Example 12.2.2. Clearly the ambipolar diffusion process can control the evolution of a dense core for a long time before free-fall collapse begins.

Numerical Simulations of Protostellar Evolution

To investigate the nature of the gravitational collapse of a cloud in detail, we must solve the magnetohydrodynamic equations numerically. Unfortunately, limits in computing power and numerical methods still necessitate making numerous and significant simplifying assumptions. These numerical models do exhibit many of the characteristics that were illustrated by our crude analytical studies, but other important aspects of the collapse become apparent that were not contained in the physics that has already been discussed.⁸

Consider a spherical cloud of approximately $1 M_{\odot}$ and solar composition that is supercritical. Initially the early stages of the free-fall collapse are nearly isothermal because light near the center of the collapse can travel significant distances before being absorbed by dust. Owing to an initial slight increase in density toward the center of the cloud, the free-fall timescale is shorter near the center and the density increases more rapidly there (inside-out collapse). When the density of the material near the center of the collapse region reaches approximately $10^{-10} \text{ kg m}^{-3}$, the region becomes optically thick and the collapse becomes more adiabatic. The opacity of the cloud at this point is primarily due to the presence of dust.

The increased pressure that occurs when the collapse becomes adiabatic substantially slows the rate of collapse near the core. At this point the central region is nearly in hydrostatic equilibrium with a radius of approximately 5 AU. It is this central object that is referred to as a protostar.

One observable consequence of the cloud becoming optically thick is that the gravitational potential energy being released during the collapse is converted into heat and then radiated away in the infrared as blackbody radiation. By computing the rate of energy release (the luminosity) and the radius of the cloud where the optical depth is $\tau = 2/3$, the effective temperature may be determined using Eq. (3.17). (At this point in its evolution, the optical depth is determined by the dust, and so the photosphere is a dust photosphere.)

With the identification of a photosphere, it becomes possible to plot the location of the simulated cloud on the H–R diagram as a function of time. Curves that depict the life histories of stars on the H–R diagram are known as **evolutionary tracks**. Figure 12.9 shows theoretical evolutionary tracks of 0.05, 0.1, 0.5, 1, 2, and $10 M_{\odot}$ clouds computed by one research group through the protostar phase. As the collapse continues to accelerate during the early stages, the luminosity of the protostar increases along with its effective temperature.

Above the developing protostellar core, material is still in free-fall. When the infalling material meets the nearly hydrostatic core, a shock wave develops where the speed of the material exceeds the local sound speed (the material is supersonic). It is at this shock front that the infalling material loses a significant fraction of its kinetic energy in the form of heat that “powers” the cloud and produces much of its luminosity.

When the temperature reaches approximately 1000 K, the dust within the developing protostar begins to vaporize and the opacity drops. This means that the radius where $\tau = 2/3$ is substantially reduced, approaching the surface of the hydrostatic core. Since the luminosity remains high during this phase, a corresponding increase in the effective temperature must occur.

⁸Some of the first calculations of protostellar collapse were performed by Richard Larson in 1969. His pioneering work neglected the complicated physics associated with rotation, turbulence, and magnetic fields but did include thermodynamics, radiative transfer, and other important physical processes.

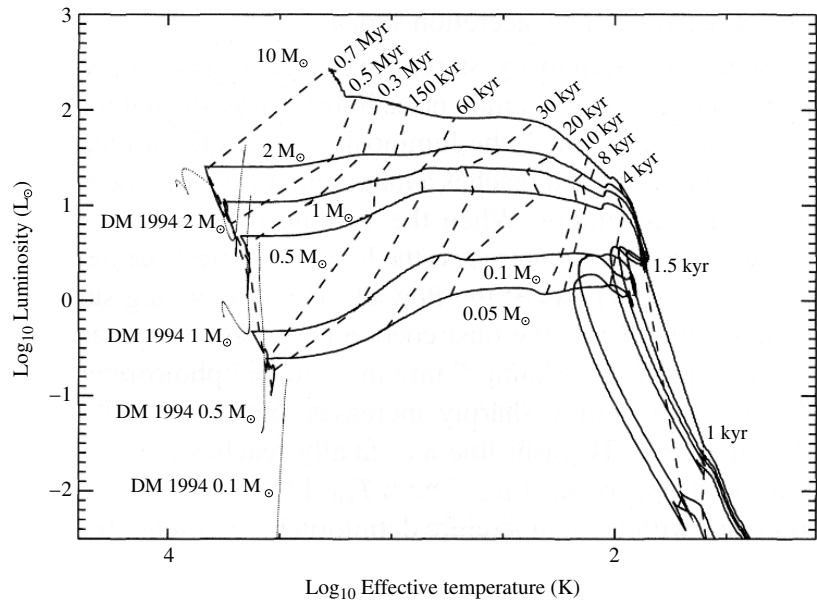


FIGURE 12.9 Theoretical evolutionary tracks of the gravitational collapse of 0.05, 0.1, 0.5, 1, 2, and 10 $M_{\odot}$ clouds through the protostar phase (solid lines). The dashed lines show the times since collapse began. The light dotted lines are pre-main-sequence evolutionary tracks of 0.1, 0.5, 1, and 2 $M_{\odot}$ stars from D'Antona and Mazzitelli, *Ap. J. Suppl.*, 90, 457, 1994. Note that the horizontal axis is plotted with effective temperature increasing to the left, as is characteristic of all H–R diagrams. (Figure adapted from Wuchterl and Tscharnuter, *Astron. Astrophys.*, 398, 1081, 2003.)

As the overlying material continues to fall onto the hydrostatic core, the temperature of the core slowly increases. Eventually the temperature becomes high enough (approximately 2000 K) to cause the molecular hydrogen to dissociate into individual atoms. This process absorbs energy that would otherwise provide a pressure gradient sufficient to maintain hydrostatic equilibrium. As a result, the core becomes dynamically unstable and a second collapse occurs.⁹ After the core radius has decreased to a value about 30% larger than the present size of the Sun, hydrostatic equilibrium is re-established. At this point, the core mass is still much less than its final value, implying that **accretion** is still ongoing.

After the core collapse, a second shock front is established as the envelope continues to accrete infalling material. When the nearly flat, roughly constant luminosity part of the evolutionary track is reached in Fig. 12.9, accretion has settled into a quasi-steady main accretion phase. At about the same time, temperatures in the deep interior of the protostar have increased enough that deuterium (^2H) begins to burn (Eq. 10.38), producing up to 60% of the luminosity of the 1 $M_{\odot}$ protostar. Note that this reaction is favored over the first step in the PP I chain because it has a fairly large cross section, $\sigma(E)$, at low temperatures.

With only a finite amount of mass available from the original cloud, and with only a limited amount of deuterium available to burn, the luminosity must eventually decrease. When deuterium burn-out occurs, the evolutionary track bends sharply downward and the

⁹Dynamical instabilities will be discussed further in Chapter 14 in connection with pulsating stars.

effective temperature decreases slightly. The evolution has now reached a quasi-static pre-main-sequence phase that will be discussed in the next section.

The theoretical scenario just described leads to the possibility of observational verification. Since it is expected that the collapse should occur deep within a molecular cloud, the protostar itself would likely be shielded from direct view by a cocoon of dust. Consequently, any observational evidence of the collapse would be in the form of small infrared sources embedded within dense cores or Bok globules. The detection of protostellar collapse is made more difficult by the relatively small value for the free-fall time, meaning that protostars are fairly short-lived objects.

The search for protostars is under way in infrared and millimeter wavelengths, and a number of strong candidates have been identified, including B335, a Bok globule in the constellation of Aquila, L1527 in Taurus, and numerous objects in the Orion Nebula. B335 is probably the best-studied case and is almost a perfect test of the theory of protostellar collapse since it seems to have very little turbulence or rotation.

Some astronomers believe that by studying the details of the infrared spectra of these sources, they have been able to identify possible spectral signatures of infalling dust and gas around the embedded infrared objects. These tell-tale features involve Doppler-shifted sub-structures in the profiles of spectral lines. For an optically thick line, a central absorption feature is often visible (see Fig. 12.10). The source of the absorption feature is cool material between the observer and the source of the line (the hotter central region). The broad wings of the line result from Doppler-shifted light coming from infalling gas. The blueshifted wing is from infalling gas on the far side of the cloud (therefore moving toward the observer), and the redshifted wing is from infalling gas on the near side of the cloud. Infall has been identified in starless dense cores as well.¹⁰

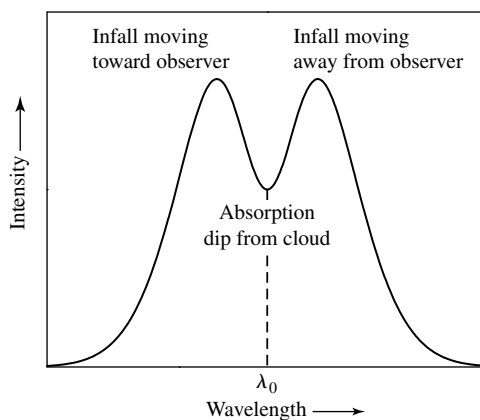


FIGURE 12.10 A line profile of a spherical, infalling cloud. The wings are Doppler shifted due to infalling material. The central absorption is produced by intervening material far from the central collapse. The redshifted wing arises from material in front of the central region moving away from the observer, and the blueshifted wing is due to material in the back of the cloud moving toward the observer.

¹⁰Another example of line profile signatures indicating mass motions is discussed in the next section; see Fig. 12.17.

12.3 ■ PRE-MAIN-SEQUENCE EVOLUTION

As we discovered in the last section, once the collapse of a molecular cloud has begun, it is characterized by the free-fall timescale given by Eq. (12.26). With the formation of a quasi-static protostar, the rate of evolution becomes controlled by the rate at which the star can thermally adjust to the collapse. This is just the Kelvin–Helmholtz timescale discussed in Example 10.3.1 (see Eq. 10.24); the gravitational potential energy liberated by the collapse is released over time and is the source of the object’s luminosity. Since $t_{\text{KH}} \gg t_{\text{ff}}$, protostellar evolution proceeds at a much slower rate than free-fall collapse. For instance, a $1 M_{\odot}$ star requires almost 40 Myr to contract quasi-statically to its main-sequence structure.

The Hayashi Track

With the steadily increasing effective temperature of the protostar, the opacity of the outer layers becomes dominated by the H^- ion, the extra electrons coming from the partial ionization of some of the heavier elements in the gas that have lower ionization potentials. As with the envelope of the main-sequence Sun, this large opacity contribution causes the envelope of a contracting protostar to become convective. In fact, in some cases the convection zone extends all the way to the center of the star. In 1961, C. Hayashi demonstrated that because of the constraints convection puts on the structure of a star, a deep convective envelope limits its quasi-static evolutionary path to a line that is nearly vertical in the H–R diagram. Consequently, as the protostar collapse slows, its luminosity decreases while its effective temperature increases slightly. It is this evolution along the **Hayashi track** that appears as the downward turn at the end of the evolutionary tracks shown in Fig. 12.9.

The Hayashi track actually represents a boundary between “allowed” hydrostatic stellar models and those that are “forbidden.” To the right of the Hayashi track, there is no mechanism that can adequately transport the luminosity out of the star at those low effective temperatures; hence no stable stars can exist there. To the left of the Hayashi track, convection and/or radiation is responsible for the necessary energy transport. Note that this distinction between allowed and forbidden models is not in conflict with the free-fall evolution of collapsing gas clouds found to the right of the Hayashi track since those objects are far from being in hydrostatic equilibrium.

Classical Calculations of Pre-Main-Sequence Evolution

In 1965, before detailed protostellar collapse calculations were performed, Icko Iben, Jr. computed the final stages of collapse onto the main sequence for stars of various masses. In each case he started his models on the Hayashi track. All of those models neglected the effects of rotation, magnetic fields, and mass loss. Since that time, significant improvements have been made in our understanding of the physical processes involved in stellar structure and evolution, including refined nuclear reaction rates, new opacities, and the inclusion of mass loss or accretion. Some modern evolutionary calculations have also included the effects of rotation.¹¹ The **pre-main-sequence evolutionary tracks** for a sequence of masses

¹¹ Some calculations have also begun considering the effects of magnetic fields, but the results presented in this text do not include those recent preliminary results.

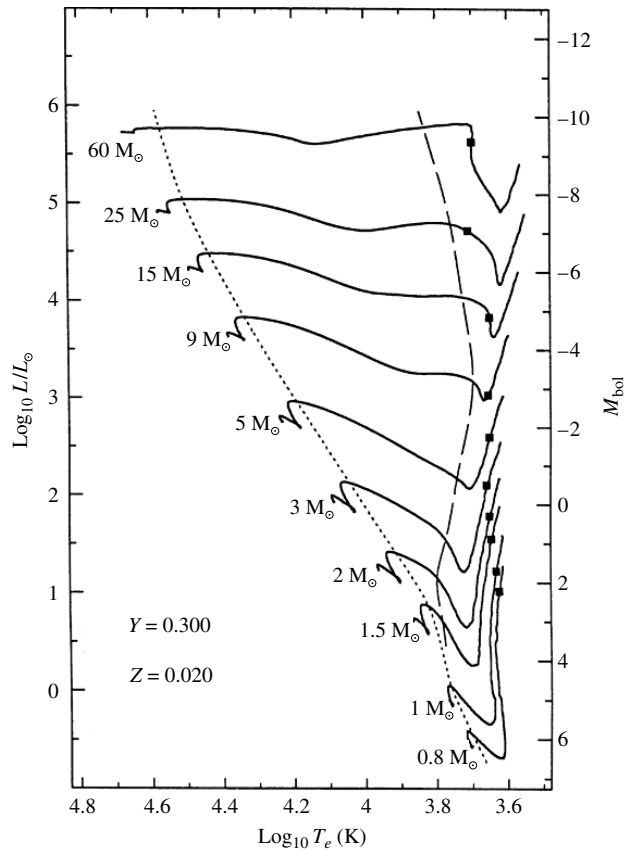


FIGURE 12.11 Classical pre-main-sequence evolutionary tracks computed for stars of various masses with the composition $X = 0.68$, $Y = 0.30$, and $Z = 0.02$. The direction of evolution on each track is generally from low effective temperature to high effective temperature (right to left). The mass of each model is indicated beside its evolutionary track. The square on each track indicates the onset of deuterium burning in these calculations. The long-dash line represents the point on each track where convection in the envelope stops and the envelope becomes purely radiative. The short-dash line marks the onset of convection in the core of the star. Contraction times for each track are given in Table 12.1. (Figure adapted from Bernasconi and Maeder, *Astron. Astrophys.*, 307, 829, 1996.)

computed with state-of-the-art physics are shown in Fig. 12.11, and the total time for each evolutionary track is given in Table 12.1.

Consider the pre-main-sequence evolution of a $1 M_{\odot}$ star, beginning on the Hayashi track. With the high H^{-} opacity near the surface, the star is completely convective during approximately the first one million years of the collapse. In these models, deuterium burning also occurs during this early period of collapse, beginning at the square indicated on the evolutionary tracks in Fig. 12.11.¹² However, since ${}^2_1\text{H}$ is not very abundant, the nuclear

¹²Note that since these calculations did not include the formation of the protostar from the direct collapse of the cloud as was done for the tracks in Fig. 12.9, there is a fundamental inconsistency between when deuterium burning occurs in the two sets of calculations.

TABLE 12.1 Pre-main-sequence contraction times for the classical models presented in Fig. 12.11. (Data from Bernasconi and Maeder, *Astron. Astrophys.*, 307, 829, 1996.)

Initial Mass ($M_{\odot}$)	Contraction Time (Myr)
60	0.0282
25	0.0708
15	0.117
9	0.288
5	1.15
3	7.24
2	23.4
1.5	35.4
1	38.9
0.8	68.4

reactions have little effect on the overall collapse; they simply slow the rate of collapse slightly.

As the central temperature continues to rise, increasing levels of ionization decrease the opacity in that region (see Fig. 9.10) and a radiative core develops, progressively encompassing more and more of the star's mass. At the point of minimum luminosity in the tracks following the descent along the Hayashi track, the existence of the radiative core allows energy to escape into the convective envelope more readily, causing the luminosity of the star to increase again. Also, as required by Eq. (3.17), the effective temperature continues to increase, since the star is still shrinking.

At about the time that the luminosity begins to increase again, the temperature near the center has become high enough for nuclear reactions to begin in earnest, although not yet at their equilibrium rates. Initially, the first two steps of the PP I chain [the conversion of ${}^1_1\text{H}$ to ${}^3_2\text{He}$; Eqs. (10.37) and (10.38)] and the CNO reactions that turn ${}^{12}_6\text{C}$ into ${}^{14}_7\text{N}$ [Eqs. (10.48–10.50)] dominate the nuclear energy production. With time, these reactions provide an increasingly larger fraction of the luminosity, while the energy production due to gravitational collapse makes less of a contribution to L .

Due to the onset of the highly temperature-dependent CNO reactions, a steep temperature gradient is established in the core, and some convection again develops in that region. At the local maximum in the luminosity on the H–R diagram near the short dashed line, the rate of nuclear energy production has become so great that the central core is forced to expand somewhat, causing the gravitational energy term in Eq. (10.36) to become negative [recall that $\epsilon = \epsilon_{\text{nuclear}} + \epsilon_{\text{gravity}}$; see Eq. (10.102)]. This effect is apparent at the surface as the total luminosity decreases toward its main-sequence value, accompanied by a decrease in the effective temperature.

When the ${}^{12}_6\text{C}$ is finally exhausted, the core completes its readjustment to nuclear burning, reaching a sufficiently high temperature for the remainder of the PP I chain to become important. At the same time, with the establishment of a stable energy source, the gravitational energy term becomes insignificant and the star finally settles onto the main sequence. It is worth noting that the time required for a $1 M_{\odot}$ star to reach the main sequence, according to the detailed numerical model just described, is not very different from the crude estimate of the Kelvin–Helmholtz timescale performed in Example 10.3.1.

For stars with masses lower than our Sun's, the evolution is somewhat different. For stars with masses $M \lesssim 0.5 M_{\odot}$ (not shown in Fig. 12.11), the upward branch is missing just before the main sequence. This happens because the central temperature never gets hot enough to burn $^{12}_6\text{C}$ efficiently (recall that our estimates of the central pressure and temperature of the Sun in Examples 10.1.1 and 10.2.1 were roughly proportional to the mass of the star). In fact, as was mentioned in Section 10.6, if the mass of the collapsing protostar is less than approximately $0.072 M_{\odot}$, the core never gets hot enough to generate sufficient energy by nuclear reactions to stabilize the star against gravitational collapse. As a result, the stable hydrogen-burning main sequence is never obtained. This explains the lower end of the main sequence.

Another important difference exists between solar-mass stars and stars of lower mass that can reach the main sequence: Temperatures remain cool enough and the opacity stays sufficiently high in low-mass stars that a radiative core never develops. Consequently, these stars remain fully convective all the way to the main sequence.

The Formation of Brown Dwarfs

Below about $0.072 M_{\odot}$, some nuclear burning will still occur, but not at a rate necessary to form a main-sequence star. Above about $0.06 M_{\odot}$ the core temperature of the star is great enough to burn lithium, and above a mass of approximately $0.013 M_{\odot}$ deuterium burning occurs ($0.013 M_{\odot}$ is roughly thirteen times the mass of Jupiter). This last value is also in agreement with the cessation of fragmentation discussed on page 420. The objects in the range between about $0.013 M_{\odot}$ and $0.072 M_{\odot}$ are known as **brown dwarfs** and have spectral types of L and T (recall Table 8.1). The first confirmed discovery of a brown dwarf, Gliese 229B, was announced in 1995. Since that time hundreds of brown dwarfs have been detected thanks to near-infrared all-sky surveys, such as the Two Micron All Sky Survey (2MASS) and the Sloan Digital Sky Survey (SDSS). Given their very low luminosities and difficulty of detection, the number of objects found to date suggest that brown dwarfs are prevalent throughout the Milky Way Galaxy.

Massive Star Formation

For massive stars, the central temperature quickly becomes high enough to burn $^{12}_6\text{C}$ as well as convert ^1_1H into ^3_2He . This means that these stars leave the Hayashi track at higher luminosities and evolve nearly horizontally across the H–R diagram. Because of the much larger central temperatures, the full CNO cycle becomes the dominant mechanism for hydrogen burning in these main-sequence stars. Since the CNO cycle is so strongly temperature-dependent, the core remains convective even after the main sequence is reached.

Possible Modifications to the Classical Models

The general pre-main-sequence evolutionary track calculations described above contain numerous approximations, as already discussed. It is likely that rotation plays an important role, along with turbulence and magnetic fields. It is also likely that the initial environments contain inhomogeneities in cloud densities, strong stellar winds, and ionizing radiation from nearby, massive stars.

These classical models also assume initial structures that are very large, with radii that are effectively infinitely greater than their final values. Given that dense cores have dimensions on the order of 0.1 pc, the initial radii of clouds undergoing protostellar collapse must be much smaller than traditionally assumed. In addition, the assumption of pressure-free protostellar collapse may also be a poor one; more realistic calculations probably require an initial contraction that is quasi-static (after all, the dark cores are roughly in hydrostatic equilibrium).

To complicate matters further, the more massive stars also interact with infalling material in such a way that a feedback loop may develop, limiting the amount of mass that they can accrete via the classical process discussed to this point; recall the discussion of the Eddington limit (e.g. Eq. 10.6).

In light of these various complications, some astronomers have suggested that significant modifications to the classical pre-main-sequence evolutionary tracks may be required. Theoretical evolutionary sequences beginning with smaller initial radii lead to a **birth line** where protostars first become visible. This birth line places an upper limit on the observed luminosities of protostars.

In addition, some observations suggest that stars with masses greater than about $10 M_{\odot}$ or so may not form at all by the classical pre-main-sequence process described above. This apparent effect could be due to limiting feedback mechanisms, such as the high luminosity of ionizing radiation associated with high effective temperatures. Instead of the collapse of single protostellar clouds, the more massive stars may form by mergers of smaller stars in dense protostellar environments. On the other hand, some researchers have argued that the need for mergers can be avoided because rotation implies that most of the infalling mass collapses to an **accretion disk**¹³ that forms around the star. The accretion disk then feeds the growing massive star, minimizing the impact of high amounts of ionizing radiation on the infalling gas and dust.

The Zero-Age Main Sequence (ZAMS)

The diagonal line in the H–R diagram where stars of various masses first reach the main sequence and begin equilibrium hydrogen burning is known as the **zero-age main sequence** (ZAMS). Inspection of the classical results given in Table 12.1 shows that the amount of time required for stars to collapse onto the ZAMS is inversely related to mass; a $0.8 M_{\odot}$ star takes over 68 Myr to reach the ZAMS, whereas a $60 M_{\odot}$ star makes it to the ZAMS in only 28,000 years!

This inverse relationship between star-formation time and stellar mass may also signal a problem with classical pre-main-sequence evolutionary models. The reason is that if the most massive stars do indeed form first in a cluster of stars, the intense radiation that they produce would likely disperse the cloud before their low-mass siblings would ever have a chance to develop.

Clearly much work remains before we can say that pre-main-sequence stellar evolution is understood.

¹³Accretion disks will be discussed in detail in Chapter 18.

The Initial Mass Function (IMF)

From observational studies it is apparent that more low-mass than high-mass stars form when an interstellar cloud fragments. This implies that the number of stars that form per mass interval per unit volume (or per unit area in the Milky Way's disk) is strongly mass-dependent. This functional dependence is known as the **initial mass function (IMF)**. One theoretical estimate of the IMF is shown in Fig. 12.12. However, a particular IMF depends on a variety of factors, including the local environment in which a cluster of stars forms from a given cloud complex in the ISM.

As a consequence of the process of fragmentation, most stars form with relatively low mass. Given the disparity in the numbers of stars formed in different mass ranges, combined with the very different rates of evolution, it is not surprising that massive stars are extremely rare, while low-mass stars are found in abundance. Observations also suggest that although the IMF is quite uncertain below about $0.1 M_{\odot}$, rather than falling off sharply as indicated in Fig. 12.12, the curve may be fairly flat, resulting in large numbers of low-mass stars and brown dwarfs.

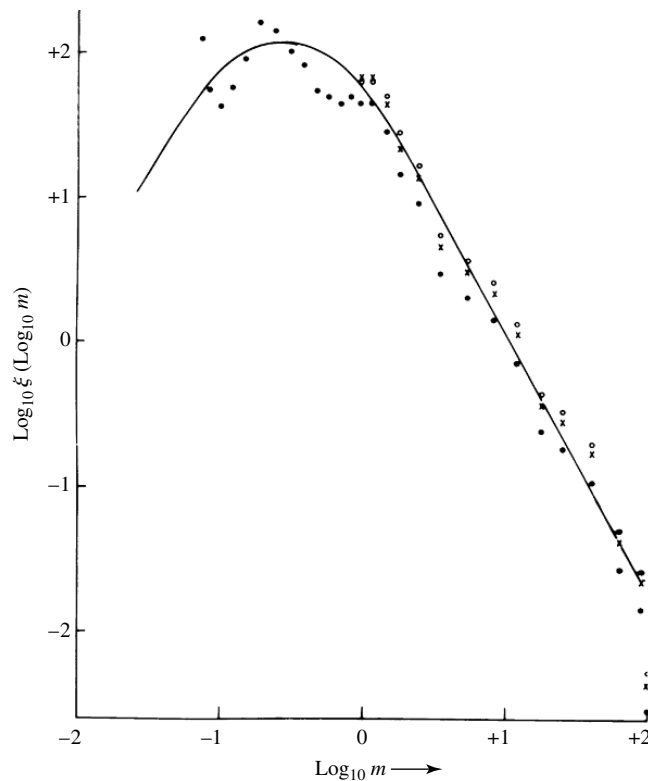


FIGURE 12.12 The initial mass function, ξ , shows the number of stars per unit area of the Milky Way's disk per unit interval of logarithmic mass that is produced in different mass intervals. The individual points represent observational data and the solid line is a theoretical estimate. Masses are in solar units. (Figure adapted from Rana, *Astron. Astrophys.*, 184, 104, 1987.)